EXTRA PROBLEMS · 1ST MIDTERM · PROBLEMS

1st Midterm: Exam-Style Problems

Exam-style problems on Units 1 to 4 (first- and second-order ODEs, linear systems), worth 5 points each, like the problems of the first partial, written for this portal and solved step by step. They are practice material, not official exercises; each one was checked with SymPy.

7 exam-style problems on Units 1 to 4 (first- and second-order ODEs, linear systems), worth 5 points each. The first partials of 2019 to 2025 are 2 or 3 problems in 60 to 90 minutes; the Exam simulator draws one ODE problem and one linear-system problem.

1st Midterm · Unit 1 · First-order modelling · 5 points · 30 min · Standard exam level

P01 — Two brine tanks in cascade: separation for the first tank, integrating factor for the second, the equal-rate case and the peak of salt

Two well-stirred tanks are connected in series. Tank A holds $V_A = 100$ L of brine containing 20 kg of dissolved salt; tank B holds V_B litres of fresh water. At $t = 0$ three pumps are switched on, all with the same flow rate $r = 10$ L/min: fresh water enters A, the mixture of A flows into B, and the mixture of B is drained to the outside. Time is measured in minutes. Let $x(t)$ and $y(t)$ be the masses of salt (kg) in A and in B, and write $k_A = r/V_A$, $k_B = r/V_B$ for the rate constants. Part (c) uses $V_B = 100$ L and part (d) uses $V_B = 50$ L.

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknowns $x(t), y(t)$ (kg), independent variable $t \geq 0$ (min).
- Hypotheses: perfect mixing (the concentration of the outflow of each tank is its mean concentration), equal inflow and outflow, so the volumes stay constant; salt is conserved (no reaction, no precipitation).
- Operator: two first-order, linear equations with constant coefficients. The equation for x is homogeneous and decoupled; the equation for y is inhomogeneous, forced by x (a triangular, one-way coupling).

Quantity	Symbol	Value	Unit
Flow rate	r	10	L/min
Volume of A	V_A	100	L
Volume of B	V_B	100 in (c), 50 in (d)	L
Initial salt in A	x_0	20	kg
Initial salt in B	$y(0)$	0	kg
Rate constants	$k_A = r/V_A$, $k_B = r/V_B$	0.1; 0.1 or 0.2	min^{-1}

Phase 2. Canonical formulation

The balance law (rate of change = rate in − rate out) for each tank, in the linear standard form $y' + p(t)\,y = q(t)$ of Topic 2:

$$x' + k_A\,x = 0, \qquad y' + k_B\,y = k_A\,x(t), \qquad x(0) = x_0, \quad y(0) = 0.$$

Part (a) · 1 point

Model and dimensions

Write the initial value problem satisfied by x and y and check its dimensions.

Step 1. Tank A

Fresh water carries no salt, so the rate in is 0. The outflow carries r L/min at concentration x/V_A kg/L:

$$\frac{dx}{dt} = 0 - r\,\frac{x}{V_A} = -k_A\,x, \qquad k_A = \frac{r}{V_A}.$$

Step 2. Tank B

Its inflow is the outflow of A, with concentration x/V_A; its outflow has concentration y/V_B:

$$\frac{dy}{dt} = r\,\frac{x}{V_A} - r\,\frac{y}{V_B} = k_A\,x - k_B\,y, \qquad k_B = \frac{r}{V_B}.$$

The initial data are $x(0) = x_0 = 20$ kg and $y(0) = 0$ (fresh water).

Step 3. Dimensions

$[r] = \text{L min}^{-1}$, $[V] = \text{L}$, so $[k_A] = [k_B] = \text{min}^{-1}$. Then $[k_A x] = \text{min}^{-1}\cdot\text{kg} = \text{kg/min} = [dx/dt]$, and the same for every term of the second equation: both equations are dimensionally homogeneous. The exponents that will appear, $k_A t$ and $k_B t$, are dimensionless.

Numerically $k_A = 10/100 = 0.1$ min^{-1}; $k_B = 10/100 = 0.1$ min^{-1} for $V_B = 100$ L and $k_B = 10/50 = 0.2$ min^{-1} for $V_B = 50$ L.

KEY RESULTS

- $x' = -\dfrac{r}{V_A}x,\ y' = \dfrac{r}{V_A}x - \dfrac{r}{V_B}y,\ x(0) = 20\ \text{kg}, y(0) = 0.$
- $k_A = k_B = 0.1\ \text{min}^{-1}$ in (c); $k_A = 0.1, k_B = 0.2\ \text{min}^{-1}$ in (d).
- Every term has units kg/min.

Part (b) · 1.5 points

Solution for different rate constants

Solve for $x(t)$ by separation of variables and, for $k_A \neq k_B$, solve for $y(t)$ with an integrating factor.

Step 1. Tank A by separation

$x \equiv 0$ is not the solution ($x_0 \neq 0$). For $x > 0$, separate: $\dfrac{dx}{x} = -k_A\,dt$. Integrate from 0 to t:

$$\int_{x_0}^{x(t)} \frac{d\xi}{\xi} = \big[\ln \xi\big]_{x_0}^{x(t)} = \ln \frac{x(t)}{x_0}, \qquad \int_0^t (-k_A)\,ds = -k_A\,t.$$

Hence $\ln(x/x_0) = -k_A t$ and

$$x(t) = x_0\,e^{-k_A t}.$$

Step 2. Integrating factor for B

Substitute $x(t)$: $y' + k_B y = k_A x_0 e^{-k_A t}$. With $p = k_B$ the integrating factor (Notes, p. 3) is $\mu(t) = e^{\int_0^t k_B\,ds} = e^{k_B t}$, with $\mu' = k_B \mu$. By the product rule, $(\mu y)' = \mu y' + \mu' y = e^{k_B t}(y' + k_B y)$, so

$$\frac{d}{dt}\big(e^{k_B t} y\big) = k_A x_0\,e^{(k_B - k_A)t}.$$

Step 3. Integration

For $k_B \neq k_A$ an antiderivative of $e^{(k_B-k_A)s}$ is $\dfrac{e^{(k_B-k_A)s}}{k_B-k_A}$. Integrating from 0 to t and using $y(0)=0$ (Barrow's rule on both sides):

$$e^{k_B t}y(t) - e^{0}\,y(0) = k_A x_0 \left[\frac{e^{(k_B-k_A)s}}{k_B-k_A}\right]_0^t = \frac{k_A x_0}{k_B-k_A}\Big(e^{(k_B-k_A)t}-1\Big).$$

Multiplying by $e^{-k_B t}$:

$$y(t) = \frac{k_A\,x_0}{k_B-k_A}\Big(e^{-k_A t}-e^{-k_B t}\Big).$$

Step 4. Check

$y(0)=\frac{k_A x_0}{k_B-k_A}(1-1)=0$. Differentiating, $y'=\frac{k_A x_0}{k_B-k_A}(-k_A e^{-k_A t}+k_B e^{-k_B t})$ and $k_A x - k_B y = \frac{k_A x_0}{k_B-k_A}\left[(k_B-k_A)e^{-k_A t} - k_B e^{-k_A t} + k_B e^{-k_B t}\right]$, which is the same expression. The quotient $\frac{e^{-k_A t}-e^{-k_B t}}{k_B-k_A}$ does not change when k_A and k_B are exchanged (numerator and denominator both change sign), and it is positive for $t>0$: $y(t)>0$ whichever tank is faster.

KEY RESULTS

- $x(t) = x_0\,e^{-k_A t}$.
- $y(t) = \dfrac{k_A\,x_0}{k_B-k_A}\big(e^{-k_A t}-e^{-k_B t}\big)$, $k_A \neq k_B$.

Part (c) · 1.5 points

Equal volumes: the resonant case

Take $V_B = 100$ L. Solve the equation for y directly, show that the result is the limit of (b) as $k_B \to k_A$, and find the time and value of the largest amount of salt in B.

Step 1. Direct solution

With $k_A = k_B = k = 0.1\ \text{min}^{-1}$, Step 2 of (b) gives $\frac{d}{dt}(e^{kt}y) = k x_0 e^{0} = k x_0$. Integrating from 0 to t: $e^{kt}y(t) - y(0) = k x_0\,[s]_0^t = k x_0\,t$, so

$$y(t) = k\,x_0\,t\,e^{-kt} = 0.1\cdot 20\,t\,e^{-t/10} = 2\,t\,e^{-t/10}\ \text{kg}.$$

The forcing kx_0e^{-kt} decays at exactly the rate of the free response e^{-kt} of tank B: the exponential guess fails and a factor t appears, the first-order analogue of resonance.

Step 2. Limit of the general formula

Fix $k_A = k$. In $y = kx_0\,\dfrac{e^{-kt} - e^{-k_Bt}}{k_B - k}$ numerator and denominator vanish at $k_B = k$. By L'Hôpital's rule in the variable k_B, with $\frac{\partial}{\partial k_B}e^{-k_Bt} = -t\,e^{-k_Bt}$ (chain rule):

$$\lim_{k_B\to k}\frac{e^{-kt} - e^{-k_Bt}}{k_B - k} = \lim_{k_B\to k}\frac{t\,e^{-k_Bt}}{1} = t\,e^{-kt},$$

so $y \to kx_0\,t\,e^{-kt}$, the result of Step 1. The two cases join continuously.

Step 3. Maximum

By the product rule, $y'(t) = kx_0\big(e^{-kt} + t(-k)e^{-kt}\big) = kx_0\,e^{-kt}(1 - kt)$. It is positive for $t < 1/k$ and negative for $t > 1/k$, so y has its maximum at $t^* = 1/k = 10$ min:

$$y_{\max} = kx_0\cdot\frac{1}{k}\,e^{-1} = \frac{x_0}{e} = \frac{20}{e} \approx 7.358\ \text{kg},$$

a concentration of $7.358/100 \approx 0.0736$ kg/L. At that moment tank A holds $x(10) = 20e^{-1}$ kg as well.

KEY RESULTS

- $y(t) = k\,x_0\,t\,e^{-kt} = 2\,t\,e^{-t/10}$ kg.
- Maximum at $t^* = 1/k = 10$ min, $y_{\max} = x_0/e = 20/e \approx 7.358$ kg (concentration 0.0736 kg/L).

Part (d) · 1 point

Smaller second tank, peak and long-time behaviour

Take $V_B = 50$ L. Find the time and value of the largest amount of salt in B, compare the concentrations of both tanks then, give the limit $t \to \infty$ and the total salt that leaves the system.

Step 1. The formula

$k_A = 0.1$, $k_B = 0.2\ \text{min}^{-1}$, so $\frac{k_Ax_0}{k_B - k_A} = \frac{0.1\cdot 20}{0.1} = 20$ kg and $y(t) = 20\big(e^{-t/10} - e^{-t/5}\big)$.

Step 2. Maximum

$y'(t) = 20\big(-\frac{1}{10}e^{-t/10} + \frac{1}{5}e^{-t/5}\big)$. With $u = e^{-t/10} \in (0,1]$: $y' = 20\big(-\frac{u}{10} + \frac{u^2}{5}\big) = \frac{20u}{10}(2u-1)$, which vanishes for $u = \frac{1}{2}$, is positive for $u > \frac{1}{2}$ (early times) and negative after. Hence $e^{-t^*/10} = \frac{1}{2}$, $t^* = 10\ln 2 \approx 6.93$ min, and

$$y_{\max} = 20\Big(\frac{1}{2} - \frac{1}{4}\Big) = 5 \text{ kg}.$$

This agrees with the general time $t^* = \dfrac{\ln(k_B/k_A)}{k_B - k_A} = \dfrac{\ln 2}{0.1}$.

Step 3. Concentrations at the peak

$x(t^*) = 20 \cdot \frac{1}{2} = 10$ kg, so A has $10/100 = 0.1$ kg/L and B has $5/50 = 0.1$ kg/L. This is general: $y' = 0$ means $k_A x = k_B y$, i.e. $\frac{x}{V_A} = \frac{y}{V_B}$. Tank B stops gaining salt exactly when its concentration equals that of the brine coming in.

Step 4. Long time and total outflow

Both exponentials tend to 0, so $x(t) \to 0$ and $y(t) \to 0$: all salt is washed out. The salt drained from B per minute is $k_B y$; with the antiderivatives $-10e^{-t/10}$ and $-5e^{-t/5}$ and Barrow's rule on $[0, T]$, $T \to \infty$:

$$\int_0^\infty 0.2 \cdot 20\big(e^{-t/10} - e^{-t/5}\big)\,dt = 4\Big[-10e^{-t/10} + 5e^{-t/5}\Big]_0^\infty = 4\big(0 - (-10+5)\big) = 20 \text{ kg},$$

the whole initial mass x_0, as conservation of salt requires.

KEY RESULTS

- $y(t) = 20\big(e^{-t/10} - e^{-t/5}\big)$ kg.
- $t^* = 10\ln 2 \approx 6.93$ min, $y_{\max} = 5$ kg; at t^* both concentrations equal 0.1 kg/L.
- $x, y \to 0$; total outflow of salt $\int_0^\infty k_B y\,dt = 20$ kg.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $t \to 0^+$: $y \approx k_A x_0\, t$ (B starts filling at the rate A loses salt, $k_A x_0 = 2$ kg/min). $t \to \infty$: $y \sim \frac{k_A x_0}{k_B - k_A} e^{-k_A t}$ when $k_A < k_B$, so the slower tank sets the decay. $V_B \to 0$ ($k_B \to \infty$): $y \to 0$, B becomes a pipe.
- **Checks:** both formulas satisfy the ODEs and $y(0) = 0$ by substitution; the equal-rate case is the limit of the general one; $\int_0^\infty k_B y\,dt = x_0$. Dimensions: $k\,x_0\,t$ is (min^{-1})(kg)(min) = kg, $t^* = 1/k$ and $\ln(k_B/k_A)/(k_B - k_A)$ are in minutes.

- **Common mistakes in the exam:** (1) using the volume of B in the inflow term of B (the inflow concentration is x/V_A); (2) applying the formula of (b) with $k_A = k_B$, which divides by zero, instead of noticing the factor t; (3) dropping the lower limit in the integrating-factor integral, which leaves $y(0) \neq 0$; (4) reporting the maximum time without checking the sign of y', or confusing the maximum of mass with the maximum of concentration.

`1st Midterm · Unit 2 · First-order qualitative analysis · 5 points · 30 min · Conceptual / fast`

P02 — Logistic population with constant harvesting: equilibria against the harvest rate, the critical rate 1/4, extinction times by partial fractions and by an arctangent

A fish population $P(t)$ grows logistically with intrinsic rate $r > 0$ and carrying capacity $K > 0$ and is fished at a constant rate $H \geq 0$ (individuals per unit time), independent of the size of the stock:

$$\frac{dP}{dt} = rP\Big(1 - \frac{P}{K}\Big) - H.$$

With $p = P/K$, $\tau = rt$ and $h = H/(rK)$ the model becomes, writing again t for τ and $'$ for $\mathrm{d}/\mathrm{d}t$,

$$p' = p(1-p) - h, \qquad h \geq 0.$$

The model is used only while $p \geq 0$; if p reaches 0 the population is extinct (collapse). Part (c) uses $h = 2/9$ and part (d) uses $h = 1/2$.

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknown $p(t) \geq 0$ (fraction of the carrying capacity), independent variable t (dimensionless time, unit $1/r$); parameter $h \geq 0$ (dimensionless harvest).
- Operator: first order, **nonlinear** (quadratic in p), **autonomous** (f does not depend on t). $f(p) = p(1-p) - h$ is a polynomial, so f and f' are continuous and Picard's theorem gives a unique local solution for every $p(0)$; distinct solutions never cross, and equilibria split the line into invariant intervals.
- Separable: $\dfrac{dp}{f(p)} = dt$ wherever $f(p) \neq 0$.

Phase 2. Canonical formulation

$$p' = f(p), \qquad f(p) = -\big(p^2 - p + h\big) = -\Big[\Big(p - \frac{1}{2}\Big)^2 + h - \frac{1}{4}\Big].$$

The completed square already shows that the sign of $h - \frac{1}{4}$ decides everything.

Part (a) · 1 point

Equilibria and the critical harvesting rate

Find the equilibria as functions of h, the critical rate h_c and its meaning.

Step 1. Equilibria

$f(p) = 0 \iff p^2 - p + h = 0$, with discriminant $1 - 4h$:

$$p_{\pm} = \frac{1 \pm \sqrt{1 - 4h}}{2} \quad (h < \tfrac{1}{4}), \qquad p = \tfrac{1}{2} \quad (h = \tfrac{1}{4}), \qquad \text{none} \quad (h > \tfrac{1}{4}).$$

For $0 \le h < \frac{1}{4}$: $0 \le p_- < \frac{1}{2} < p_+ \le 1$, with $p_- = 0, p_+ = 1$ when $h = 0$ (the logistic equation). As h grows the two equilibria approach each other, merge at $\frac{1}{2}$ for $h = \frac{1}{4}$ and disappear.

Step 2. Critical rate and meaning

$h_c = \frac{1}{4}$. The natural growth $g(p) = p(1-p)$ has $g'(p) = 1 - 2p = 0$ at $p = \frac{1}{2}$, with $g(\frac{1}{2}) = \frac{1}{4}$, its maximum. In original variables $h = H/(rK)$, so

$$H_c = \frac{rK}{4},$$

the largest growth the stock can produce, reached at $P = K/2$. Any constant catch above it exceeds what the population can regenerate at any size (the "maximum sustainable yield"). The merging and disappearance of two equilibria as a parameter crosses a value is called a saddle-node bifurcation (extension: a name beyond the course; the course only needs the phase lines of (b)).

KEY RESULTS

- $p_{\pm} = \dfrac{1 \pm \sqrt{1 - 4h}}{2}$ for $h < \frac{1}{4}$; one equilibrium $p = \frac{1}{2}$ for $h = \frac{1}{4}$; none for $h > \frac{1}{4}$.
- $h_c = \frac{1}{4}$, i.e. $H_c = \dfrac{rK}{4}$, the maximum natural growth rate (reached at $P = K/2$).

Part (b) · 1.5 points

Phase lines in the three regimes

Describe the phase line and classify the equilibria for $h < h_c$, $h = h_c$, $h > h_c$, and say which initial populations survive.

Step 1. Case $0 \le h < \frac{1}{4}$

$f(p) = -(p - p_-)(p - p_+)$: $f < 0$ for $p < p_-$, $f > 0$ on (p_-, p_+), $f < 0$ for $p > p_+$. Phase line (arrows of p'):

$$0 \quad \longleftarrow \quad p_- \quad \longrightarrow \quad p_+ \quad \longleftarrow$$

With $f'(p) = 1 - 2p$ (Notes, p. 3, linear stability test):

$$f'(p_\pm) = 1 - \left(1 \pm \sqrt{1 - 4h}\right) = \mp\sqrt{1 - 4h},$$

so $f'(p_-) > 0$: **unstable**; $f'(p_+) < 0$: **asymptotically stable**. Populations with $p(0) > p_-$ tend to p_+; those with $p(0) < p_-$ decrease and reach 0 (p_- is a survival threshold). For $h = 0, p_- = 0$ and nothing collapses.

Step 2. Case $h = \frac{1}{4}$

$f(p) = -(p - \frac{1}{2})^2 \le 0$, with $f'(\frac{1}{2}) = 0$: the linear test is inconclusive and the sign of f decides. $p' < 0$ on both sides:

$$\longleftarrow \quad \tfrac{1}{2} \quad \longleftarrow$$

From above, p decreases to $\frac{1}{2}$ (attracting side); from below, p moves away towards 0 (repelling side). The equilibrium is **semi-stable**: it is not stable, since an arbitrarily small loss pushes the stock into collapse.

Step 3. Case $h > \frac{1}{4}$

$f(p) = -(p - \frac{1}{2})^2 - (h - \frac{1}{4}) \le -(h - \frac{1}{4}) < 0$ for every p. There is no equilibrium and every solution decreases at a rate at least $h - \frac{1}{4}$, so from any $p(0)$ it reaches 0 before the time $p(0)/(h - \frac{1}{4})$: collapse in finite time for every initial stock.

KEY RESULTS

- $h < \frac{1}{4}$: p_- unstable ($f'(p_-) = \sqrt{1 - 4h} > 0$), p_+ asymptotically stable ($f'(p_+) = -\sqrt{1 - 4h} < 0$). Survival iff $p(0) \ge p_-$ (with $p(0) = p_-$ the equilibrium itself).
- $h = \frac{1}{4}$: $p = \frac{1}{2}$ semi-stable ($f = -(p - \frac{1}{2})^2 \le 0$): attracts from above, repels below. Survival iff $p(0) \ge \frac{1}{2}$.
- $h > \frac{1}{4}$: $f \le \frac{1}{4} - h < 0$ everywhere, every solution decreases and collapses in finite time.

Part (c) · 1.5 points

Explicit solutions for h = 2/9

Solve with $p(0) = 1/2$ and with $p(0) = 1/4$; in the second case find the time of collapse.

Step 1. Factorisation and partial fractions

With $h = \frac{2}{9}$: $\sqrt{1 - \frac{8}{9}} = \frac{1}{3}$, $p_\pm = \frac{1 \pm 1/3}{2}$, i.e. $p_- = \frac{1}{3}$, $p_+ = \frac{2}{3}$, and $f(p) = -(p - \frac{1}{3})(p - \frac{2}{3})$. Write

$$\frac{1}{(p - \frac{1}{3})(p - \frac{2}{3})} = \frac{A}{p - \frac{1}{3}} + \frac{B}{p - \frac{2}{3}} \iff 1 = A\Big(p - \frac{2}{3}\Big) + B\Big(p - \frac{1}{3}\Big).$$

At $p = \frac{1}{3}$: $1 = -\frac{1}{3}A$, $A = -3$. At $p = \frac{2}{3}$: $1 = \frac{1}{3}B$, $B = 3$.

Step 2. Separation and integration

For $p \ne \frac{1}{3}, \frac{2}{3}$: $\dfrac{dp}{(p - \frac{1}{3})(p - \frac{2}{3})} = -dt$. By Step 1, $\displaystyle\int \Big(-\frac{3}{p - \frac{1}{3}} + \frac{3}{p - \frac{2}{3}} \Big) dp = -\int dt$. With the antiderivatives $\int \frac{dp}{p - c} = \ln|p - c|$:

$$3 \ln \left| p - \frac{2}{3} \right| - 3 \ln \left| p - \frac{1}{3} \right| = -t + C \implies \ln \left| \frac{p - \frac{2}{3}}{p - \frac{1}{3}} \right| = -\frac{t}{3} + \frac{C}{3}.$$

Exponentiating and absorbing the sign into the constant: $\dfrac{p - \frac{2}{3}}{p - \frac{1}{3}} = K\, e^{-t/3}$, with $K = \dfrac{p(0) - \frac{2}{3}}{p(0) - \frac{1}{3}}$ from $t = 0$. The constant solutions $p \equiv \frac{1}{3}$ and $p \equiv \frac{2}{3}$, lost when dividing by $(p - \frac{1}{3})(p - \frac{2}{3})$, correspond to $K = \infty$ and $K = 0$.

Step 3. Solving for p

Write $E = Ke^{-t/3}$: $p - \frac{2}{3} = E\,p - \frac{E}{3}$, so $p(1-E) = \frac{2}{3} - \frac{E}{3}$ and

$$p(t) = \frac{2 - K\,e^{-t/3}}{3\big(1 - K\,e^{-t/3}\big)}.$$

Step 4. The two initial data

- $p(0) = \frac{1}{2}$: $K = \dfrac{-1/6}{1/6} = -1,\ p(t) = \dfrac{2 + e^{-t/3}}{3(1 + e^{-t/3})}$. Check $p(0) = \frac{3}{6} = \frac{1}{2}$. As $t \to \infty$, $e^{-t/3} \to 0$ and $p \to \frac{2}{3} = p_+$: the stock recovers to the stable equilibrium, as the phase line predicts.
- $p(0) = \frac{1}{4}$ (below the threshold $p_- = \frac{1}{3}$): $K = \dfrac{\frac{1}{4} - \frac{2}{3}}{\frac{1}{4} - \frac{1}{3}} = \dfrac{-5/12}{-1/12} = 5$. Multiplying numerator and denominator by $e^{t/3}$:

$$p(t) = \frac{2e^{t/3} - 5}{3\big(e^{t/3} - 5\big)}.$$

Check $p(0) = \frac{-3}{-12} = \frac{1}{4}$. $p = 0$ when $2e^{t/3} = 5$:

$$T_0 = 3\ln\frac{5}{2} \approx 2.749.$$

The formula has a pole at $t = 3\ln 5 \approx 4.83 > T_0$, but the model has already stopped: the population is extinct at T_0. In original time, $t_{\text{real}} = T_0/r$.

KEY RESULTS

- Equilibria $p_- = \frac{1}{3}$ (unstable), $p_+ = \frac{2}{3}$ (stable). General solution $\dfrac{p - \frac{2}{3}}{p - \frac{1}{3}} = K\,e^{-t/3}, p = \dfrac{2 - Ke^{-t/3}}{3(1 - Ke^{-t/3})}$.
- $p(0) = \frac{1}{2}$: $K = -1, p(t) = \dfrac{2 + e^{-t/3}}{3(1 + e^{-t/3})} \to \frac{2}{3}$.
- $p(0) = \frac{1}{4}$: $K = 5, p(t) = \dfrac{2e^{t/3} - 5}{3(e^{t/3} - 5)}$, collapse at $T_0 = 3\ln\frac{5}{2} \approx 2.749$.

Part (d) · 1 point

Collapse time for h = 1/2

Take $h = 1/2$ *and* $p(0) = 1$. *Find the collapse time* T *and* $p(t)$ *on* $[0, T]$.

Step 1. Antiderivative

$p' = -\left[(p - \frac{1}{2})^2 + \frac{1}{4}\right] < 0$. With $s = 2p - 1$, $ds = 2\,dp$ and $(p - \frac{1}{2})^2 + \frac{1}{4} = \frac{s^2+1}{4}$:

$$\int \frac{dp}{(p - \frac{1}{2})^2 + \frac{1}{4}} = \int \frac{ds/2}{(s^2+1)/4} = 2\int \frac{ds}{1 + s^2} = 2\arctan s = 2\arctan(2p - 1).$$

Step 2. Time to collapse

Separating, $dt = -\dfrac{dp}{(p - \frac{1}{2})^2 + \frac{1}{4}}$. As t runs from 0 to T, p runs from 1 down to 0, so by Barrow's rule

$$T = \int_1^0 \frac{-dp}{(p - \frac{1}{2})^2 + \frac{1}{4}} = \Big[2\arctan(2p - 1)\Big]_0^1 = 2\arctan 1 - 2\arctan(-1) = 2 \cdot \frac{\pi}{4} + 2 \cdot \frac{\pi}{4} = \pi.$$

The same computation from a general $p(0)$ gives $T = 2\arctan(2p(0) - 1) - 2\arctan(-1) = 2\arctan(2p(0) - 1) + \frac{\pi}{2}$; for example $p(0) = \frac{1}{2}$ gives $\frac{\pi}{2}$.

Step 3. The solution

Integrating from 0 to t: $2\arctan(2p - 1) - 2\arctan(1) = -t$, so $\arctan(2p - 1) = \frac{\pi}{4} - \frac{t}{2}$ and

$$p(t) = \frac{1}{2} + \frac{1}{2}\tan\left(\frac{\pi}{4} - \frac{t}{2}\right), \qquad 0 \le t \le \pi.$$

Check: $p(0) = \frac{1}{2} + \frac{1}{2}\tan\frac{\pi}{4} = 1$; $p(\pi) = \frac{1}{2} + \frac{1}{2}\tan(-\frac{\pi}{4}) = 0$. By the chain rule $p' = \frac{1}{2}\sec^2(\frac{\pi}{4} - \frac{t}{2}) \cdot (-\frac{1}{2}) = -\frac{1}{4}(1 + \tan^2)$, and $-\left[(p - \frac{1}{2})^2 + \frac{1}{4}\right] = -\frac{1}{4}\tan^2 - \frac{1}{4}$: the same. In original time the collapse happens at $t_{\text{real}} = \pi/r$.

KEY RESULTS

- $T = \displaystyle\int_0^1 \frac{dp}{(p - \frac{1}{2})^2 + \frac{1}{4}} = \Big[2\arctan(2p - 1)\Big]_0^1 = \pi.$
- $p(t) = \dfrac{1}{2} + \dfrac{1}{2}\tan\left(\dfrac{\pi}{4} - \dfrac{t}{2}\right), 0 \le t \le \pi.$
- For a general start $p(0) \in [0, 1]$: $T = 2\arctan(2p(0) - 1) + \dfrac{\pi}{2}.$

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $h = 0$ recovers the logistic equation ($p_- = 0, p_+ = 1$). For $h > \frac{1}{4}$ and $p(0) = 1$, with $\omega = \sqrt{h - \frac{1}{4}}$ the same arctangent integral gives $T = \frac{2}{\omega} \arctan \frac{1}{2\omega}$ ($= \pi$ for $\omega = \frac{1}{2}$): $T \to \infty$ as $h \to \frac{1}{4}^+$ (the stock lingers near $p = \frac{1}{2}$, where p' is almost zero) and $T \to 0$ as $h \to \infty$.
- **Checks:** every explicit solution was substituted back into the equation and satisfies its initial value; the solutions agree with the phase lines of (b) (monotone, no crossing of equilibria). Dimensions: $h = H/(rK)$ is (individuals/time)/((1/time)(individuals)), dimensionless; real times are the dimensionless ones divided by r.
- **Common mistakes in the exam:** (1) applying the linear test $f'(p^*)$ at $h = \frac{1}{4}$, where it gives 0, and concluding "stable"; (2) forgetting that below p_- the population collapses even for small harvesting; (3) reading the pole of the formula of (c) as the time of extinction instead of the first zero of p; (4) losing the factor 2 from $ds = 2\,dp$ in the arctangent integral.

`1st Midterm · Unit 3 · Reduction of order · 5 points · 30 min · Standard exam level`

P03 — A variable-coefficient equation with the solution x: reduction of order to x e^x, Abel's identity against the direct Wronskian, variation of parameters for the forcing x^4 and an IVP

Consider, for $x > 0$, the equation

$$x^2 y'' - x(x+2)\,y' + (x+2)\,y = g_0(x).$$

The homogeneous equation ($g_0 = 0$) has a solution that is a polynomial of degree one. In parts (c) and (d) the right-hand side is $g_0(x) = x^4$, and the initial data of (d) are

$$y(1) = \frac{1}{2}, \qquad y'(1) = -\frac{1}{2}.$$

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknown $y(x)$, independent variable x on the interval $(0, \infty)$.
- Operator: second order, **linear**, with **variable** coefficients; homogeneous in (a) and (b), inhomogeneous in (c) and (d).
- The leading coefficient x^2 vanishes at $x = 0$: $x = 0$ is a singular point, so all statements are made on $x > 0$, where the coefficients of the standard form are continuous and the existence and uniqueness theorem holds (Topic 3, Concept - Existence and Uniqueness Theorem for Second-Order ODEs).

Phase 2. Canonical formulation

Dividing by $x^2 \neq 0$, with $\frac{x+2}{x^2} = \frac{1}{x} + \frac{2}{x^2}$, the standard form $y'' + p(x)y' + q(x)y = g(x)$ is

$$y'' - \Big(1 + \frac{2}{x}\Big)y' + \Big(\frac{1}{x} + \frac{2}{x^2}\Big)y = \frac{g_0(x)}{x^2}, \qquad p(x) = -\Big(1 + \frac{2}{x}\Big), \quad g(x) = x^2 \text{ for } g_0 = x^4.$$

Part (a) · 1.5 points

First solution and reduction of order

Find the polynomial solution y_1 of degree one and, with $y = v\,y_1$, a second solution y_2 linearly independent of y_1.

Step 1. Polynomial solution

With $y = \alpha x + \beta$: $y' = \alpha$, $y'' = 0$, and

$$-x(x+2)\alpha + (x+2)(\alpha x + \beta) = (x+2)\big(-\alpha x + \alpha x + \beta\big) = \beta\,(x+2).$$

This vanishes for all x only if $\beta = 0$. So $y_1 = x$ (taking $\alpha = 1$).

Step 2. Substitution y = v x

By the product rule, $y = vx$ gives $y' = v'x + v$ and $y'' = v''x + v' + v' = v''x + 2v'$. Substituting:

$$x^2\big(xv'' + 2v'\big) - x(x+2)\big(xv' + v\big) + (x+2)\,xv.$$

Expanding term by term: $x^3v'' + 2x^2v' - (x^3 + 2x^2)v' - (x^2 + 2x)v + (x^2 + 2x)v = x^3v'' - x^3v'$. The terms in v cancel because y_1 solves the equation, as they must. Hence

$$x^3\big(v'' - v'\big) = 0 \iff v'' = v' \qquad (x > 0).$$

Step 3. First-order equation for w = v'

$w' = w$ is separable: $\frac{dw}{w} = dx$, $\ln|w| = x + C_1$, $w = Ce^x$. Integrating once more, $v = \int Ce^x\,dx = Ce^x + D$. Then $y = vx = C\,x\,e^x + D\,x$. The part Dx is the old solution; the new one is

$$y_2 = x\,e^x.$$

Check: $y_2' = (1+x)e^x$, $y_2'' = (2+x)e^x$; $x^2(2+x) - x(x+2)(1+x) + (x+2)x = x(x+2)\big[x - (1+x) + 1\big] = 0$, times e^x.

KEY RESULTS

- $y_1 = x$.
- $y = v\,x$ gives $x^3(v'' - v') = 0$, so $v' = Ce^x$ and $y_2 = x\,e^x$.
- General homogeneous solution $y_h = c_1 x + c_2\,x\,e^x$.

Part (b) · 1 point

Wronskian by Abel's identity and directly

Compute the Wronskian of $\{y_1, y_2\}$ from Abel's identity and from the determinant, and conclude that they form a fundamental set on $x > 0$.

Step 1. Abel's identity

With $p = -(1 + \frac{2}{x})$, $W' = -pW = \left(1 + \frac{2}{x}\right)W$. Separating, $\frac{dW}{W} = \left(1 + \frac{2}{x}\right)dx$, and with the antiderivative $x + 2\ln x$ ($x > 0$):

$$\ln|W| = x + 2\ln x + C_1 \implies W(x) = C\,x^2 e^x.$$

Step 2. Direct determinant

$$W = \det\begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} = \det\begin{pmatrix} x & x\,e^x \\ 1 & (1+x)e^x \end{pmatrix} = x(1+x)e^x - x\,e^x \cdot 1 = x^2 e^x.$$

The two results agree with $C = 1$.

Step 3. Conclusion

$x^2 e^x > 0$ for $x > 0$, so y_1, y_2 are linearly independent there and $y_h = c_1 x + c_2 x e^x$ is the general solution of the homogeneous equation (Robinson Ch. 11). The dichotomy "either $W \equiv 0$ or W never vanishes" applies on intervals where p is continuous; $W(0) = 0$ happens at the singular point $x = 0$ and is no contradiction.

KEY RESULTS

- Abel: $W = C\,x^2 e^x$.
- Determinant: $W = x(1+x)e^x - x\,e^x = x^2 e^x$, so $C = 1$.
- $W \neq 0$ on $(0, \infty)$: fundamental set; $W(0) = 0$ only at the singular point.

Part (c) · 1.5 points

Variation of parameters for the forcing x^4

With $g_0(x) = x^4$, find a particular solution by variation of parameters and write the general solution.

Step 1. Formula

For $y'' + py' + qy = g$ with fundamental set $\{y_1, y_2\}$ (Robinson Eq. 18.8, Notes, p. 8):

$$y_p = -y_1 \int \frac{y_2\, g}{W}\, dx + y_2 \int \frac{y_1\, g}{W}\, dx, \qquad g = \frac{x^4}{x^2} = x^2.$$

Step 2. The two integrands

$$\frac{y_2\, g}{W} = \frac{x\, e^{x} \cdot x^2}{x^2 e^{x}} = x, \qquad \frac{y_1\, g}{W} = \frac{x \cdot x^2}{x^2 e^{x}} = x\, e^{-x}.$$

Step 3. The integrals

$\int x\, dx = \frac{x^2}{2}$. For the second, integrate by parts with $u = x$, $dv = e^{-x}dx$, so $du = dx$, $v = -e^{-x}$:

$$\int x\, e^{-x} dx = -x\, e^{-x} - \int (-e^{-x})\, dx = -x\, e^{-x} - e^{-x} = -(x+1)e^{-x}.$$

(Integration constants are omitted: they only add multiples of y_1 and y_2.)

Step 4. Particular and general solution

$$y_p = -x \cdot \frac{x^2}{2} + x\, e^{x} \cdot \big(-(x+1)e^{-x} \big) = -\frac{x^3}{2} - x^2 - x.$$

The term $-x$ is a multiple of y_1 and can be absorbed: $y_p = -\dfrac{x^3}{2} - x^2$. Check, with $y_p' = -\frac{3}{2}x^2 - 2x$, $y_p'' = -3x - 2$:

$$x^2(-3x-2) - x(x+2)\Big(-\frac{3}{2}x^2 - 2x\Big) + (x+2)\Big(-\frac{x^3}{2} - x^2\Big)$$

The products are $x(x+2)\big(\frac{3}{2}x^2 + 2x\big) = \frac{3}{2}x^4 + 2x^3 + 3x^3 + 4x^2 = \frac{3}{2}x^4 + 5x^3 + 4x^2$ and $(x+2)\big(-\frac{x^3}{2} - x^2\big) = -\frac{1}{2}x^4 - x^3 - x^3 - 2x^2 = -\frac{1}{2}x^4 - 2x^3 - 2x^2$, so

$$= \big(-3x^3 - 2x^2\big) + \Big(\frac{3}{2}x^4 + 5x^3 + 4x^2\Big) + \Big(-\frac{1}{2}x^4 - 2x^3 - 2x^2\Big) = x^4. \ \checkmark$$

The general solution is $y = c_1 x + c_2\, x\, e^{x} - \dfrac{x^3}{2} - x^2$.

KEY RESULTS

- $\displaystyle\int \frac{y_2\, g}{W}dx = \int x\, dx = \frac{x^2}{2}$, $\displaystyle\int \frac{y_1\, g}{W}dx = \int x\, e^{-x}dx = -(x+1)e^{-x}$.
- $y_p = -\dfrac{x^3}{2} - x^2$ (after dropping the multiple $-x$ of y_1).
- $y = c_1 x + c_2 x e^{x} - \dfrac{x^3}{2} - x^2$.

Part (d) · 1 point

Initial value problem

Solve the IVP with $y(1) = \frac{1}{2}$, $y'(1) = -\frac{1}{2}$ and give the largest interval on which the solution is guaranteed to exist and be unique.

Step 1. Conditions

$y(1) = c_1 + c_2 e - \frac{1}{2} - 1 = \frac{1}{2} \Rightarrow c_1 + e\, c_2 = 2$. $y'(1) = c_1 + 2e\, c_2 - \frac{3}{2} - 2 = -\frac{1}{2} \Rightarrow c_1 + 2e\, c_2 = 3$.

Step 2. Constants

Subtracting the first equation from the second: $e\, c_2 = 1$, $c_2 = e^{-1}$; then $c_1 = 2 - 1 = 1$. The system is uniquely solvable because its matrix is $\begin{pmatrix} y_1(1) & y_2(1) \\ y_1'(1) & y_2'(1) \end{pmatrix} = \begin{pmatrix} 1 & e \\ 1 & 2e \end{pmatrix}$, whose determinant $2e - e = e$ is exactly $W(1) \neq 0$.

$$y(x) = x + x\, e^{x-1} - \frac{x^3}{2} - x^2.$$

Step 3. Interval

p, q and g of the standard form are continuous on $(0, \infty)$, which contains $x_0 = 1$; the existence and uniqueness theorem gives a unique solution on the whole of $(0, \infty)$. Nothing is guaranteed at or across $x = 0$.

KEY RESULTS

- $c_1 + e\, c_2 = 2$ and $c_1 + 2e\, c_2 = 3$, so $c_2 = e^{-1}$, $c_1 = 1$.
- $y(x) = x + x\, e^{x-1} - \dfrac{x^3}{2} - x^2$ on $(0, \infty)$.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** as $x \to \infty$ the term $x\, e^{x-1}$ dominates and $y \to +\infty$; as $x \to 0^+$, $y \to 0$ and $y'(0^+) = 1 + e^{-1}$ (finite, though the theorem does not cover $x = 0$).
- **Checks:** y_1, y_2 and y_p were substituted back; Abel's W and the determinant agree; $y(1) = 1 + 1 - \frac{1}{2} - 1 = \frac{1}{2}$ and $y'(1) = 1 + 2 - \frac{3}{2} - 2 = -\frac{1}{2}$.
- **Common mistakes in the exam:** (1) applying variation of parameters to x^4 without dividing by the leading coefficient x^2; (2) taking $p = -(x+2)$ instead of $-(x+2)/x$ in Abel's identity; (3) a sign slip in $\int x e^{-x} dx$; (4) keeping $-x$ in y_p and still reporting $c_1 = 1$: with $y_p = -\frac{x^3}{2} - x^2 - x$ the IVP gives $c_1 = 2$, and the solution $y(x)$ is the same.

1st Midterm · Unit 3 · Constant coefficients, Cauchy-Euler and forcing · 5 points · 30 min · High difficulty / trap

P04 — Inverse problem: rebuild y'' + a y' + b y = f from one homogeneous solution and one particular solution, then the resonant forcing e^{-x} sin 2x, an IVP and which forcings need the modified guess

The real constants a, b of the equation

$$y'' + a\,y' + b\,y = f(x)$$

are unknown. Two facts are known about it:

- *the function $y = e^{-x}\cos 2x$ solves the homogeneous equation ($f = 0$);*
- *when the right-hand side is $f(x) = k\,x$, for a certain constant k, the polynomial $y = 5x - 2$ is a solution.*

Afterwards the equation is driven by $f(x) = 8\,e^{-x}\sin 2x$, with initial data $y(0) = 1$, $y'(0) = -3$. Finally, consider forcings of the form $f(x) = e^{\alpha x}\cos \omega x$ or $e^{\alpha x}\sin \omega x$ with real α and $\omega \geq 0$, and in particular the purely oscillatory forcing $f(x) = \cos \omega x$.

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknowns: first the parameters a, b, k (inverse problem), then the function $y(x)$, $x \in \mathbb{R}$.
- Operator: second order, linear, constant coefficients, $L[y] = y'' + ay' + by$. Coefficients and forcings are continuous on $\mathbb{R}$, so every IVP has a unique solution on all of $\mathbb{R}$.
- Superposition: $y = y_h + y_p$, with y_h from the characteristic equation $r^2 + ar + b = 0$ and y_p from undetermined coefficients.

Phase 2. Canonical formulation

The equation is already in standard form. Writing $y = e^{-x}u$ (an exponential shift, **extension**: not in the course, it only organises the computation of (b)) turns L into an operator on u that will expose the resonance.

Part (a) · 1.5 points

Reconstruction of the equation

Find a, b and k, and write the general solution of the homogeneous equation.

Step 1. Derivatives of the given solution

Product and chain rule on $y = e^{-x}\cos 2x$:

$$y' = -e^{-x}\cos 2x - 2e^{-x}\sin 2x = e^{-x}\big(-\cos 2x - 2\sin 2x\big),$$

$$y'' = -e^{-x}\big(-\cos 2x - 2\sin 2x\big) + e^{-x}\big(2\sin 2x - 4\cos 2x\big) = e^{-x}\big(-3\cos 2x + 4\sin 2x\big).$$

Step 2. Conditions on a and b

$$L[y] = e^{-x}\Big[\big(-3 - a + b\big)\cos 2x + \big(4 - 2a\big)\sin 2x\Big] = 0 \quad \forall x.$$

$e^{-x} \neq 0$ and $\cos 2x$, $\sin 2x$ are linearly independent, so both brackets vanish: $4 - 2a = 0 \Rightarrow a = 2$, and $-3 - 2 + b = 0 \Rightarrow b = 5$.

Cross-check: $e^{-x}\cos 2x = \operatorname{Re} e^{(-1+2i)x}$ requires the root $r = -1 + 2i$; real coefficients give also $-1 - 2i$, so $r^2 + ar + b = (r + 1 - 2i)(r + 1 + 2i) = (r+1)^2 + 4 = r^2 + 2r + 5$.

Step 3. The constant k

$y = 5x - 2$: $y' = 5$, $y'' = 0$, so $L[y] = 0 + 2\cdot 5 + 5(5x - 2) = 25x$. Hence $k = 25$. Undetermined coefficients give the same: with $y_p = Ax + B$, $L[y_p] = 2A + 5Ax + 5B = 25x$, so $5A = 25$, $A = 5$, and $2A + 5B = 0$, $B = -2$. The second fact is consistent with $a = 2$, $b = 5$ (it imposes $5a = 2b$ through the constant term, which they satisfy).

Step 4. Homogeneous solution

$r^2 + 2r + 5 = 0$: $r = \dfrac{-2 \pm \sqrt{4 - 20}}{2} = -1 \pm 2i$ (Robinson Ch. 12), so

$$y_h = e^{-x}\big(c_1\cos 2x + c_2\sin 2x\big).$$

KEY RESULTS

- $a = 2, b = 5$; characteristic roots $r = -1 \pm 2i$.
- $k = 25$; $y_p = 5x - 2$ is recovered by the guess $Ax + B$.
- $y_h = e^{-x}\big(c_1\cos 2x + c_2\sin 2x\big)$.

Part (b) · 1.5 points

Resonant forcing and the modified guess

For $f(x) = 8\,e^{-x}\sin 2x$ explain why the usual guess fails and find a particular solution.

Step 1. Why the usual guess fails

The forcing is $\operatorname{Im}\left(8e^{(-1+2i)x}\right)$ and $-1+2i$ is a simple root of r^2+2r+5. The guess $e^{-x}(A\cos 2x + B\sin 2x)$ lies in y_h, so L of it is 0 for every A, B, never $8e^{-x}\sin 2x$. The rule (Robinson 14.5) multiplies the guess by x (once, because the root is simple).

Step 2. Exponential shift

Let $y = e^{-x}u$. By the product rule, $y' = e^{-x}(u' - u)$ and $y'' = -e^{-x}(u'-u) + e^{-x}(u''-u') = e^{-x}(u'' - 2u' + u)$. Then

$$L[y] = e^{-x}\Big[(u''-2u'+u) + 2(u'-u) + 5u\Big] = e^{-x}\big(u'' + 4u\big).$$

So $L[y] = 8e^{-x}\sin 2x$ becomes $u'' + 4u = 8\sin 2x$: an undamped oscillator forced exactly at its natural frequency 2.

Step 3. Coefficients

With $u = x(A\cos 2x + B\sin 2x)$:

$$u' = (A\cos 2x + B\sin 2x) + x(-2A\sin 2x + 2B\cos 2x),$$

$$u'' = 2(-2A\sin 2x + 2B\cos 2x) + x(-4A\cos 2x - 4B\sin 2x).$$

The terms with x cancel against $4u$: $x(-4A\cos 2x - 4B\sin 2x) + 4x(A\cos 2x + B\sin 2x) = 0$, so $u'' + 4u = 4B\cos 2x - 4A\sin 2x$. Matching with $8\sin 2x$: $4B = 0$ and $-4A = 8$, so $A = -2$, $B = 0$:

$$y_p = -2x\,e^{-x}\cos 2x.$$

KEY RESULTS

- $e^{-x}(A\cos 2x + B\sin 2x)$ is a homogeneous solution: L annihilates it.
- Modified guess $y_p = x\,e^{-x}(A\cos 2x + B\sin 2x)$; $L[y_p] = e^{-x}\big(4B\cos 2x - 4A\sin 2x\big)$.
- $A = -2, B = 0$: $y_p = -2x\,e^{-x}\cos 2x$.

Part (c) · 1 point

Initial value problem

Solve the IVP with $f(x) = 8e^{-x}\sin 2x$, $y(0) = 1$, $y'(0) = -3$, and describe its behaviour as $x \to \infty$.

Step 1. General solution and its derivative

$y = e^{-x}(c_1\cos 2x + c_2\sin 2x) - 2xe^{-x}\cos 2x$. Differentiating with the product rule and evaluating at $x = 0$ ($\cos 0 = 1, \sin 0 = 0$):

$$\frac{d}{dx}\Big[e^{-x}(c_1\cos 2x + c_2\sin 2x)\Big]_{x=0} = -c_1 + 2c_2, \qquad \frac{d}{dx}\Big[-2xe^{-x}\cos 2x\Big]_{x=0} = -2e^{0}\cos 0 = -2.$$

Step 2. Constants

$y(0) = c_1 = 1$; $y'(0) = -c_1 + 2c_2 - 2 = -3 \Rightarrow 2c_2 = 0, c_2 = 0$:

$$y(x) = (1 - 2x)\,e^{-x}\cos 2x.$$

Step 3. Long-time behaviour

$|y| \le |1 - 2x|e^{-x} \to 0$, because e^{-x} beats any polynomial. Here "resonance" does not mean unbounded growth: the factor x only slows the decay, since the damping ($a = 2 > 0$) sets the envelope e^{-x}.

KEY RESULTS

- $y'(0) = -c_1 + 2c_2 - 2$; $c_1 = 1, c_2 = 0$.
- $y(x) = (1 - 2x)\,e^{-x}\cos 2x \to 0$ as $x \to \infty$.

Part (d) · 1 point

Which forcings need the modified guess

For $f = e^{\alpha x}\cos\omega x$ or $e^{\alpha x}\sin\omega x$, when is the modified guess needed? For $f = \cos\omega x$, find the amplitude of the particular solution as a function of ω and the frequency that maximises it.

Step 1. The rule

The guess $e^{\alpha x}(A\cos\omega x + B\sin\omega x)$ fails only when it solves the homogeneous equation, i.e. when $\alpha + i\omega \in \{-1+2i, -1-2i\}$: $\alpha = -1$ and $\omega = 2$. For a polynomial or $e^{\alpha x}$ ($\omega = 0$) the guess would need a real root, and there is none. For $\cos\omega x$ ($\alpha = 0$) the real part 0 never equals -1: **no frequency** of a purely oscillatory forcing needs the modified guess.

Step 2. Coefficients for cos(omega x)

$y_p' = -A\omega\sin\omega x + B\omega\cos\omega x$, $y_p'' = -\omega^2 y_p$. Then

$$L[y_p] = \left[(5-\omega^2)A + 2\omega B\right]\cos\omega x + \left[(5-\omega^2)B - 2\omega A\right]\sin\omega x = \cos\omega x.$$

From the second bracket $B = \frac{2\omega}{5-\omega^2}A$ (for $\omega^2 \neq 5$); substituting into the first, $A\,\frac{(5-\omega^2)^2+4\omega^2}{5-\omega^2} = 1$. Hence $A = \frac{5-\omega^2}{D}$, $B = \frac{2\omega}{D}$ with $D = (5-\omega^2)^2 + 4\omega^2$ (the same formulas hold at $\omega^2 = 5$, solving directly).

Step 3. Amplitude and its maximum

$R = \sqrt{A^2+B^2} = \dfrac{\sqrt{(5-\omega^2)^2+4\omega^2}}{D} = \dfrac{1}{\sqrt{D}}$. Expanding, $D = \omega^4 - 10\omega^2 + 25 + 4\omega^2 = \omega^4 - 6\omega^2 + 25 = (\omega^2-3)^2 + 16 \geq 16$. D is smallest, and R largest, at $\omega^2 = 3$:

$$\omega = \sqrt{3}, \qquad R_{\max} = \frac{1}{\sqrt{16}} = \frac{1}{4}.$$

This "practical resonance" is a finite peak, not a failure of the guess: it is the damped counterpart of the undamped resonance, and it sits below the frequency 2 of the free oscillation.

KEY RESULTS

- Modified guess only for $\alpha = -1, \omega = 2$. Never for $\cos\omega x$ ($\alpha = 0$), for polynomials or for real exponentials.
- $A = \dfrac{5-\omega^2}{D}, B = \dfrac{2\omega}{D}, D = (5-\omega^2)^2 + 4\omega^2 = (\omega^2-3)^2 + 16$; amplitude $R(\omega) = \dfrac{1}{\sqrt{D}}$.
- Maximum at $\omega = \sqrt{3}$, with $R_{\max} = \dfrac{1}{4}$.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** as $x \to \infty$ every solution of the homogeneous equation decays like e^{-x}, so the long-time response to $\cos \omega x$ is y_p alone; $R(0) = \frac{1}{5}$ (static response $1/b$) and $R(\omega) \sim \omega^{-2}$ as $\omega \to \infty$.
- **Checks:** $L[e^{-x}\cos 2x] = 0$, $L[5x - 2] = 25x$, $L[-2xe^{-x}\cos 2x] = 8e^{-x}\sin 2x$ and the IVP were verified by substitution. With zero damping and forcing at the natural frequency (for example $y'' + 4y = 8\sin 2x$) the factor x would grow without bound (true resonance); here $a = 2$ gives the envelope e^{-x}, so $xe^{-x} \to 0$.
- **Common mistakes in the exam:** (1) deducing only b from the homogeneous solution and forgetting that the $\sin 2x$ coefficient fixes a; (2) trying $e^{-x}(A\cos 2x + B\sin 2x)$ for the resonant forcing and getting $0 = 8$; (3) calling $\omega = 2$ or $\omega = \sqrt{3}$ resonant for $\cos \omega x$: the guess never needs modifying there; (4) losing the term -2 from $\frac{d}{dx}(-2xe^{-x}\cos 2x)$ at $x = 0$.

1st Midterm · Unit 4 · Linear systems and phase portraits · 5 points · 30 min · Standard exam level

P05 — The system x' = A(a) x with A = (-4, a; 1, 0): type of the origin for every a, the stable node a = -3 with P, P^{-1}, P^{-1}AP, and the stable focus a = -8 in real form with its sense of rotation

For a real parameter a consider the planar system

$$\dot{\mathbf{x}} = A(a)\,\mathbf{x}, \qquad A(a) = \begin{pmatrix} -4 & a \\ 1 & 0 \end{pmatrix}, \qquad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

The type of the origin changes with a. Afterwards two values are studied in detail: $a = -3$, with initial condition $\mathbf{x}(0) = (1, 1)^T$, and $a = -8$, with initial condition $\mathbf{x}(0) = (0, 1)^T$.

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknown $\mathbf{x}(t) \in \mathbb{R}^2$, $t \in \mathbb{R}$; parameter $a \in \mathbb{R}$.
- Operator: first-order, linear, homogeneous, constant coefficients: solutions exist for all t, are unique, and $\mathbf{x} \equiv \mathbf{0}$ is always an equilibrium. It is the only one when $\det A \neq 0$.
- The type of $\mathbf{0}$ is read from $\tau = \operatorname{tr} A$, $\delta = \det A$ and $\Delta = \tau^2 - 4\delta$ (**extension**: the (τ, δ, Δ) table is derived from $\lambda^2 - \tau\lambda + \delta = 0$; Robinson Ch. 31 lists the cases by type of eigenvalue).

Phase 2. Canonical formulation

$\tau = -4 + 0 = -4$ for every a, $\delta = (-4)(0) - a \cdot 1 = -a$, $\Delta = 16 + 4a$. With an invertible matrix P of (generalised) eigenvectors, $\mathbf{x} = P\mathbf{z}$ turns the system into $\dot{\mathbf{z}} = (P^{-1}AP)\,\mathbf{z}$ with $P^{-1}AP$ diagonal, a Jordan block, or the rotation-dilation block $\begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$.

Part (a) · 1.5 points

Classification for all a

Classify the origin for every real a (table with trace, determinant and discriminant) and describe the borderline cases.

Step 1. Characteristic polynomial

$$\det(A - \lambda I) = \det \begin{pmatrix} -4 - \lambda & a \\ 1 & -\lambda \end{pmatrix} = (-4 - \lambda)(-\lambda) - a = \lambda^2 + 4\lambda - a,$$

$$\lambda_{\pm} = \frac{-4 \pm \sqrt{16 + 4a}}{2} = -2 \pm \sqrt{4 + a}.$$

Step 2. Table

Range of a	τ	$\delta = -a$	$\Delta = 16 + 4a$	**Eigenvalues**	**Origin**
$a > 0$	-4	< 0	> 16	real, opposite signs	saddle (unstable)
$a = 0$	-4	0	16	0 and -4	line of equilibria $x_1 = 0$
$-4 < a < 0$	-4	> 0	$0 < \Delta < 16$	real, both negative	stable node
$a = -4$	-4	4	0	-2 double	improper stable node
$a < -4$	-4	> 4	< 0	$-2 \pm i\sqrt{-4 - a}$	stable focus

For $-4 < a < 0$: $0 < \sqrt{4 + a} < 2$, so both $-2 \pm \sqrt{4 + a}$ are negative. For $a > 0$: $\sqrt{4 + a} > 2$, so $\lambda_+ > 0 > \lambda_-$.

Step 3. Borderline values

- $a = 0$: $A\mathbf{x} = \mathbf{0}$ means $-4x_1 = 0$ and $x_1 = 0$, so every point of the x_2-axis is an equilibrium (eigenvalue 0, eigenvector $(0, 1)^T$). For $\lambda = -4$, $(A + 4I)\mathbf{v} = \mathbf{0}$ has rows $(0, 0)$ and $(1, 4)$: $v_1 + 4v_2 = 0$, $\mathbf{v} = (-4, 1)^T$. Every other orbit is a half-line parallel to $(-4, 1)^T$ that ends on the x_2-axis as $t \to \infty$.
- $a = -4$: $A + 2I = \begin{pmatrix} -2 & -4 \\ 1 & 2 \end{pmatrix} \neq 0$; its second row gives $v_1 + 2v_2 = 0$, a single eigenvector line through $(-2, 1)^T$. The origin is an improper stable node: all orbits approach it tangent to that line.
- $a < -4$: $\rho = -2, \omega = \sqrt{-4 - a}$. Sense of rotation from the test vector $(1, 0)^T$: $A(1, 0)^T = (-4, 1)^T$, whose second component is positive on the positive x_1-axis, so orbits turn **anticlockwise** (equivalently $a_{12} = a < 0$).

Since $\tau = -4 \neq 0$ there is never a centre, and since $\tau < 0$ there is never an unstable node or focus: the only unstable case is the saddle.

KEY RESULTS

- $\lambda^2 + 4\lambda - a = 0$, $\lambda = -2 \pm \sqrt{4 + a}$.
- $a > 0$ saddle; $a = 0$ line of equilibria $x_1 = 0$; $-4 < a < 0$ stable node; $a = -4$ improper stable node, single eigenvector $(-2, 1)^T$; $a < -4$ stable focus, anticlockwise.
- Never a centre and never unstable apart from the saddle.

Part (b) · 1.5 points

The stable node a = -3

For $a = -3$ find eigenvalues and eigenvectors, build P, P^{-1}, check $P^{-1}AP$, write the general solution, solve with $\mathbf{x}(0) = (1, 1)^T$ and describe the phase portrait.

Step 1. Eigenvectors

$A = \begin{pmatrix} -4 & -3 \\ 1 & 0 \end{pmatrix}$, $\lambda^2 + 4\lambda + 3 = (\lambda + 1)(\lambda + 3)$.

- $\lambda_1 = -1$: $A + I = \begin{pmatrix} -3 & -3 \\ 1 & 1 \end{pmatrix}$. Row 1: $-3v_1 - 3v_2 = 0$; row 2: $v_1 + v_2 = 0$ (the same line). $\mathbf{v}_1 = (1, -1)^T$.
- $\lambda_2 = -3$: $A + 3I = \begin{pmatrix} -1 & -3 \\ 1 & 3 \end{pmatrix}$. Row 1: $-v_1 - 3v_2 = 0$; row 2: $v_1 + 3v_2 = 0$. $\mathbf{v}_2 = (3, -1)^T$.

Step 2. Change of basis

$$P = \begin{pmatrix} 1 & 3 \\ -1 & -1 \end{pmatrix}, \qquad \det P = (1)(-1) - (3)(-1) = 2, \qquad P^{-1} = \frac{1}{2}\begin{pmatrix} -1 & -3 \\ 1 & 1 \end{pmatrix}.$$

Column by column, $AP = [A\mathbf{v}_1 \mid A\mathbf{v}_2]$ with $A\mathbf{v}_1 = (-4 + 3,\ 1)^T = (-1, 1)^T = -\mathbf{v}_1$ and $A\mathbf{v}_2 = (-12 + 3,\ 3)^T = (-9, 3)^T = -3\mathbf{v}_2$:

$$AP = \begin{pmatrix} -1 & -9 \\ 1 & 3 \end{pmatrix}, \qquad P^{-1}AP = \frac{1}{2}\begin{pmatrix} (-1)(-1) + (-3)(1) & (-1)(-9) + (-3)(3) \\ (1)(-1) + (1)(1) & (1)(-9) + (1)(3) \end{pmatrix} = \frac{1}{2}\begin{pmatrix} -2 & 0 \\ 0 & -6 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -3 \end{pmatrix}.$$

Step 3. Solution

In $\mathbf{z} = P^{-1}\mathbf{x}$: $\dot z_1 = -z_1$, $\dot z_2 = -3z_2$, so $z_1 = c_1 e^{-t}$, $z_2 = c_2 e^{-3t}$ (each separable: $\frac{dz_i}{z_i} = \lambda_i\, dt$). Back with $\mathbf{x} = P\mathbf{z}$:

$$\mathbf{x}(t) = c_1 e^{-t}\begin{pmatrix} 1 \\ -1 \end{pmatrix} + c_2 e^{-3t}\begin{pmatrix} 3 \\ -1 \end{pmatrix}.$$

For $\mathbf{x}(0) = (1, 1)^T$: $(c_1, c_2)^T = P^{-1}(1, 1)^T = \frac{1}{2}(-1 - 3,\ 1 + 1)^T = (-2, 1)^T$, so

$$\mathbf{x}(t) = \begin{pmatrix} -2e^{-t} + 3e^{-3t} \\ 2e^{-t} - e^{-3t} \end{pmatrix}, \qquad \mathbf{x}(0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Step 4. Phase portrait in words

Both eigenvalues are negative: **stable node**. The eigenlines through $(1,-1)^T$ and $(3,-1)^T$ are invariant, with arrows towards the origin. In the $\mathbf{z}$-plane the orbits are $z_2 = K|z_1|^3$ (exponent $\lambda_2/\lambda_1 = 3$), tangent to the z_1-axis at the origin. Back in $\mathbf{x}$: every orbit not on the fast line enters the origin tangent to the **slow** eigenvector $\mathbf{v}_1 = (1,-1)^T$ ($|\lambda_1| = 1$), and far from the origin ($t \to -\infty$) it is parallel to the fast one $\mathbf{v}_2 = (3,-1)^T$. The orbit of the IVP behaves like $-2e^{-t}\mathbf{v}_1 = e^{-t}(-2,2)^T$ for large t: it enters from the second quadrant along $x_2 = -x_1$. Nullclines: $\dot{x}_1 = 0$ on $x_2 = -\frac{4}{3}x_1$, $\dot{x}_2 = 0$ on $x_1 = 0$ (there the motion is horizontal, $\dot{x}_1 = -3x_2$).

KEY RESULTS

- $\lambda_1 = -1, \mathbf{v}_1 = (1,-1)^T$; $\lambda_2 = -3, \mathbf{v}_2 = (3,-1)^T$.
- $P = \begin{pmatrix} 1 & 3 \\ -1 & -1 \end{pmatrix}$, $\det P = 2$, $P^{-1} = \frac{1}{2}\begin{pmatrix} -1 & -3 \\ 1 & 1 \end{pmatrix}$, $P^{-1}AP = \operatorname{diag}(-1,-3)$.
- $\mathbf{x}(t) = \left(-2e^{-t} + 3e^{-3t},\ 2e^{-t} - e^{-3t}\right)^T$; stable node, tangent to $\mathbf{v}_1$ at the origin.

Part (c) · 2 points

The stable focus a = -8

For $a = -8$ find the complex eigenvalues and an eigenvector $\mathbf{v} = \mathbf{u} + i\mathbf{w}$, build $Q = [\mathbf{u} \mid \mathbf{w}]$, Q^{-1} and check $Q^{-1}AQ$; write the real general solution, solve with $\mathbf{x}(0) = (0,1)^T$ and find the sense of rotation.

Step 1. Eigenvalues and eigenvector

$\lambda^2 + 4\lambda + 8 = 0$: $\lambda = \frac{-4 \pm \sqrt{16-32}}{2} = -2 \pm 2i$. For $\lambda = -2 + 2i$:

$$A - \lambda I = \begin{pmatrix} -2 - 2i & -8 \\ 1 & 2 - 2i \end{pmatrix}.$$

Row 2: $v_1 + (2-2i)v_2 = 0$; with $v_2 = 1$, $v_1 = -2 + 2i$. Row 1 checks: $(-2-2i)(-2+2i) - 8 = (4 - 4i + 4i - 4i^2) - 8 = 8 - 8 = 0$. So $\mathbf{v} = (-2+2i, 1)^T = \mathbf{u} + i\mathbf{w}$ with $\mathbf{u} = (-2,1)^T$, $\mathbf{w} = (2,0)^T$.

Step 2. Action of A on u and w

$A\mathbf{u} = (8-8,\ -2)^T = (0,-2)^T = \rho\mathbf{u} - \omega\mathbf{w} = (4,-2)^T - (4,0)^T$. $A\mathbf{w} = (-8,\ 2)^T = \omega\mathbf{u} + \rho\mathbf{w} = (-4,2)^T + (-4,0)^T$. Both agree with Robinson Eq. 29.3.

Step 3. Change of basis

$$Q = \begin{pmatrix} -2 & 2 \\ 1 & 0 \end{pmatrix}, \qquad \det Q = (-2)(0) - (2)(1) = -2, \qquad Q^{-1} = \frac{1}{-2}\begin{pmatrix} 0 & -2 \\ -1 & -2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & 1 \end{pmatrix}.$$

$AQ = [A\mathbf{u} \mid A\mathbf{w}] = \begin{pmatrix} 0 & -8 \\ -2 & 2 \end{pmatrix}$ and

$$Q^{-1}AQ = \begin{pmatrix} 0\cdot 0 + 1\cdot(-2) & 0\cdot(-8) + 1\cdot 2 \\ \frac{1}{2}\cdot 0 + 1\cdot(-2) & \frac{1}{2}\cdot(-8) + 1\cdot 2 \end{pmatrix} = \begin{pmatrix} -2 & 2 \\ -2 & -2 \end{pmatrix} = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}.$$

Step 4. Real general solution and the IVP

In $\mathbf{z}$: $\dot z_1 = -2z_1 + 2z_2$, $\dot z_2 = -2z_1 - 2z_2$; with $r^2 = z_1^2 + z_2^2$: $r\dot r = z_1\dot z_1 + z_2\dot z_2 = z_1(-2z_1 + 2z_2) + z_2(-2z_1 - 2z_2) = -2r^2$ and $r^2\dot\theta = z_1\dot z_2 - z_2\dot z_1 = z_1(-2z_1 - 2z_2) - z_2(-2z_1 + 2z_2) = -2r^2$, i.e. $\dot r = -2r$, $\dot\theta = -2$, so $r = r_0 e^{-2t}$, $\theta = \theta_0 - 2t$ (Barrow: $[e^{2s}r(s)]_0^t = 0$). Back in $\mathbf{x}$:

$$\mathbf{x}(t) = e^{-2t}\Big[(a_1\cos 2t + b_1\sin 2t)\,\mathbf{u} + (b_1\cos 2t - a_1\sin 2t)\,\mathbf{w}\Big], \qquad (a_1,b_1)^T = Q^{-1}\mathbf{x}(0).$$

For $\mathbf{x}(0) = (0,1)^T$: $(a_1,b_1)^T = (0\cdot 0 + 1\cdot 1,\ \frac{1}{2}\cdot 0 + 1\cdot 1)^T = (1,1)^T$. Then

$$\mathbf{x}(t) = e^{-2t}\Big[(\cos 2t + \sin 2t)\begin{pmatrix} -2 \\ 1 \end{pmatrix} + (\cos 2t - \sin 2t)\begin{pmatrix} 2 \\ 0 \end{pmatrix}\Big] = e^{-2t}\begin{pmatrix} -4\sin 2t \\ \cos 2t + \sin 2t \end{pmatrix}.$$

Check: $\mathbf{x}(0) = (0,1)^T$. By the product rule $\dot x_1 = e^{-2t}(8\sin 2t - 8\cos 2t)$, and $-4x_1 - 8x_2 = e^{-2t}\big(16\sin 2t - 8\cos 2t - 8\sin 2t\big) = e^{-2t}(8\sin 2t - 8\cos 2t)$; $\dot x_2 = e^{-2t}\big(-2\cos 2t - 2\sin 2t - 2\sin 2t + 2\cos 2t\big) = -4e^{-2t}\sin 2t = x_1$.

Step 5. Rotation and portrait

At $(1,0)^T$, $\dot{\mathbf{x}} = A(1,0)^T = (-4,1)^T$: the velocity on the positive x_1-axis points upward, so orbits turn **anticlockwise** in the $\mathbf{x}$-plane. In the $\mathbf{z}$-plane they turn clockwise ($\dot\theta = -2$); the sense flips because $\det Q = -2 < 0$. The origin is a **stable focus**: spirals shrinking by the factor $e^{2\pi\rho/\omega} = e^{-2\pi} \approx 0.0019$ each turn (one turn every π time units). The IVP orbit starts at $(0,1)^T$, moves into the half-plane $x_1 < 0$ and crosses the x_2-axis again at $t = \pi/2$, at $(0, -e^{-\pi})^T$.

KEY RESULTS

- $\lambda = -2 \pm 2i$ ($\rho = -2, \omega = 2$); $\mathbf{v} = (-2+2i,\ 1)^T$, $\mathbf{u} = (-2, 1)^T$, $\mathbf{w} = (2, 0)^T$.
- $Q = \begin{pmatrix} -2 & 2 \\ 1 & 0 \end{pmatrix}$, $\det Q = -2$, $Q^{-1} = \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & 1 \end{pmatrix}$, $Q^{-1}AQ = \begin{pmatrix} -2 & 2 \\ -2 & -2 \end{pmatrix}$.
- $\mathbf{x}(t) = e^{-2t}\big(-4\sin 2t,\ \cos 2t + \sin 2t\big)^T$; stable focus, anticlockwise in $\mathbf{x}$ (clockwise in $\mathbf{z}$), radius factor $e^{-2\pi}$ per turn.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** for every $a < 0$ all solutions tend to $\mathbf{0}$ as $t \to \infty$ (asymptotic stability); for $a = 0$ they tend to points of the x_2-axis; for $a > 0$ only the orbits on the stable eigenline reach the origin. As $a \to -4^{\pm}$ the node's two eigenlines merge (from $a > -4$) or the spiral's rotation rate $\omega = \sqrt{-4-a}$ goes to 0 (from $a < -4$): the improper node is the common boundary.
- **Checks:** $P^{-1}AP$ and $Q^{-1}AQ$ were multiplied out; both IVP solutions satisfy the system and the initial data; $\operatorname{tr}(P^{-1}AP) = -4 = \tau$ and $\det(P^{-1}AP) = 3 = \delta$ for $a = -3$, and the same for $a = -8$ (-4 and 8).
- **Common mistakes in the exam:** (1) calling $a = 0$ a node: the origin is not isolated; (2) ordering $Q = [\mathbf{w} \mid \mathbf{u}]$ and then expecting $\begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$ (that order gives the transpose); (3) reading the sense of rotation from the $\mathbf{z}$-plane, which is always clockwise, instead of from a test vector in $\mathbf{x}$; (4) making the node's orbits tangent to the fast eigenvector at the origin.

1st Midterm · Unit 4 · Linear systems and phase portraits · 5 points · 30 min · High difficulty / trap

P06 — Inverse problem: rebuild the matrix of a spiral sink from one real solution, solve an IVP and measure the decay per turn

The real 2×2 matrix A of the linear system $\mathbf{x}' = A\mathbf{x}$, $\mathbf{x} = (x, y)^T$, is unknown. The only information available is that the system has the real solution

$$\mathbf{x}_1(t) = e^{-t} \begin{pmatrix} \cos 2t - \sin 2t \\ \cos 2t - 2\sin 2t \end{pmatrix}.$$

(a) Reconstruct A. Show that its eigenvalues are $-1 \pm 2i$, find a complex eigenvector, and check that $P^{-1}AP$ is the real canonical form, with P the matrix whose columns are the real and imaginary parts of that eigenvector, normalised so that the given solution is $\mathrm{Re}(e^{\lambda t}\boldsymbol{\eta})$.

(b) Find a second real solution, prove that the two solutions form a fundamental set, write the real general solution and solve the initial value problem $\mathbf{x}(0) = (1, 0)^T$.

(c) Classify the origin. Do the trajectories spiral in or out? Is the rotation clockwise or counter-clockwise? Justify with a test vector.

(d) (extension: the quadratic form V is not in the course; it follows from the canonical form.) Let $V(x, y) = 5x^2 - 6xy + 2y^2$. Show that $V(\mathbf{x}) = |P^{-1}\mathbf{x}|^2$ and that $\frac{d}{dt}V(\mathbf{x}(t)) = -2V(\mathbf{x}(t))$ along every solution. For the solution of (b), find the time needed for $\sqrt{V}$ to halve, and the first time $t > 0$ at which the solution is back on the positive x axis, with the ratio of its distances to the origin there and at $t = 0$. Is the Euclidean distance $|\mathbf{x}(t)|$ decreasing for every solution?

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknowns: the four real entries of A; then the solutions $\mathbf{x}(t) \in \mathbb{R}^2$, $t \in \mathbb{R}$.
- Operator: first-order, linear, homogeneous, constant coefficients, planar. Solutions exist for all t and are unique for given $\mathbf{x}(0)$ (existence and uniqueness for linear systems).
- Datum: one solution $\mathbf{x}_1$. Since $\mathbf{x}_1'(t) = A\mathbf{x}_1(t)$ for every t, evaluating at two instants where $\mathbf{x}_1$ takes independent values gives A on a basis, hence A.
- Course convention (Robinson Ch. 29): if $\lambda = \rho + i\omega$ ($\omega > 0$) has eigenvector $\boldsymbol{\eta} = \mathbf{u} + i\mathbf{w}$, the real solutions are $e^{\rho t}(\cos \omega t\, \mathbf{u} - \sin \omega t\, \mathbf{w})$ and $e^{\rho t}(\sin \omega t\, \mathbf{u} + \cos \omega t\, \mathbf{w})$.

Quantity	Value
$\mathbf{x}_1(0)$	$(1, 1)^T$
$\mathbf{x}_1(\pi/4)$	$e^{-\pi/4}(-1, -2)^T$

Quantity	Value
Initial condition (b)	$\mathbf{x}(0) = (1, 0)^T$

Phase 2. Canonical formulation

Matrix form $\mathbf{x}' = A\mathbf{x}$. The target canonical form ("frame" of the problem) is $\mathbf{z}' = J\mathbf{z}$ with $\mathbf{x} = P\mathbf{z}$, $P = [\mathbf{u} \mid \mathbf{w}]$ and $J = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$, so that $A = PJP^{-1}$.

Part (a) · 1.5 points

Reconstruct A, its eigenvalues and eigenvector, and check the canonical form

Step 1. Derivative of the given solution

Write $c = \cos 2t$, $s = \sin 2t$. By the chain rule $\frac{d}{dt}\cos 2t = -2\sin 2t$, $\frac{d}{dt}\sin 2t = 2\cos 2t$, and by the product rule $\frac{d}{dt}\big(e^{-t}g(t)\big) = -e^{-t}g(t) + e^{-t}g'(t)$:

$$\mathbf{x}_1'(t) = -e^{-t}\begin{pmatrix} c - s \\ c - 2s \end{pmatrix} + e^{-t}\begin{pmatrix} -2s - 2c \\ -2s - 4c \end{pmatrix} = e^{-t}\begin{pmatrix} -3c - s \\ -5c \end{pmatrix}$$

Step 2. Two instants give A on a basis

At $t = 0$ ($c = 1, s = 0$): $\mathbf{x}_1(0) = (1, 1)^T$, $\mathbf{x}_1'(0) = (-3, -5)^T$. At $t = \pi/4$ ($c = 0, s = 1$): $\mathbf{x}_1(\pi/4) = e^{-\pi/4}(-1, -2)^T$, $\mathbf{x}_1'(\pi/4) = e^{-\pi/4}(-1, 0)^T$. Dividing the second relation by $-e^{-\pi/4}$:

$$A\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -3 \\ -5 \end{pmatrix}, \qquad A\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

With $P = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ this reads $AP = \begin{pmatrix} -3 & 1 \\ -5 & 0 \end{pmatrix}$. $\det P = 1 \cdot 2 - 1 \cdot 1 = 1 \neq 0$, and $P^{-1} = \frac{1}{1}\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$, so $A = (AP)P^{-1}$ entry by entry:

$$\begin{aligned} A_{11} &= (-3)(2) + (1)(-1) = -7, & A_{12} &= (-3)(-1) + (1)(1) = 4, \\ A_{21} &= (-5)(2) + (0)(-1) = -10, & A_{22} &= (-5)(-1) + (0)(1) = 5. \end{aligned}$$

$$A = \begin{pmatrix} -7 & 4 \\ -10 & 5 \end{pmatrix}$$

Step 3. Eigenvalues

$$p(\lambda) = \det(A - \lambda I) = (-7-\lambda)(5-\lambda) - (4)(-10) = \lambda^2 + 2\lambda - 35 + 40 = \lambda^2 + 2\lambda + 5$$

$\lambda = \frac{-2 \pm \sqrt{4-20}}{2} = -1 \pm 2i$, so $\rho = -1, \omega = 2$. Check: $\operatorname{tr} A = -2 = 2\rho$, $\det A = -35 + 40 = 5 = \rho^2 + \omega^2$.

Step 4. Eigenvector from $(A - \lambda I)\boldsymbol{\eta} = \mathbf{0}$

For $\lambda = -1 + 2i$, row 1: $(-7 - (-1+2i))\eta_1 + 4\eta_2 = 0$, i.e. $(-6 - 2i)\eta_1 + 4\eta_2 = 0$, so $\eta_2 = \frac{3+i}{2}\eta_1$. Taking $\eta_1 = 1 + i$: $\eta_2 = \frac{(3+i)(1+i)}{2} = \frac{3+3i+i-1}{2} = 1 + 2i$. Row 2 as a check: $-10(1+i) + (5 - (-1+2i))(1+2i) = -10 - 10i + (6-2i)(1+2i) = -10 - 10i + 6 + 12i - 2i + 4 = 0$. Hence

$$\boldsymbol{\eta} = \begin{pmatrix} 1+i \\ 1+2i \end{pmatrix} = \underbrace{\begin{pmatrix} 1 \\ 1 \end{pmatrix}}_{\mathbf{u}} + i \underbrace{\begin{pmatrix} 1 \\ 2 \end{pmatrix}}_{\mathbf{w}}$$

and indeed $\mathbf{x}_1 = e^{-t}(\cos 2t\,\mathbf{u} - \sin 2t\,\mathbf{w})$: the given curve is the real part of $e^{\lambda t}\boldsymbol{\eta}$.

Step 5. Frame check $P^{-1}AP = J$

$P = [\mathbf{u} \mid \mathbf{w}]$ is the matrix of Step 2. With $AP = \begin{pmatrix} -3 & 1 \\ -5 & 0 \end{pmatrix}$:

$$\begin{aligned} (P^{-1}AP)_{11} &= 2(-3) + (-1)(-5) = -1, & (P^{-1}AP)_{12} &= 2(1) + (-1)(0) = 2, \\ (P^{-1}AP)_{21} &= (-1)(-3) + 1(-5) = -2, & (P^{-1}AP)_{22} &= (-1)(1) + 1(0) = -1. \end{aligned}$$

$$P^{-1}AP = \begin{pmatrix} -1 & 2 \\ -2 & -1 \end{pmatrix} = \begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix} = J$$

KEY RESULTS

- $A = \begin{pmatrix} -7 & 4 \\ -10 & 5 \end{pmatrix}, \operatorname{tr} A = -2, \det A = 5, p(\lambda) = \lambda^2 + 2\lambda + 5, \lambda = -1 \pm 2i.$
- $\boldsymbol{\eta} = (1+i,\ 1+2i)^T, \mathbf{u} = (1,1)^T, \mathbf{w} = (1,2)^T.$
- $P = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \det P = 1, P^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}, P^{-1}AP = \begin{pmatrix} -1 & 2 \\ -2 & -1 \end{pmatrix}.$

Part (b) · 1.5 points

Second real solution, fundamental set, general solution and IVP

Step 1. Complex solution split into real and imaginary parts

With Euler's formula $e^{(-1+2i)t} = e^{-t}(\cos 2t + i \sin 2t)$:

$$e^{\lambda t}\boldsymbol{\eta} = e^{-t}(\cos 2t + i \sin 2t)(\mathbf{u} + i\mathbf{w}) = e^{-t}\big[(\cos 2t\,\mathbf{u} - \sin 2t\,\mathbf{w}) + i(\sin 2t\,\mathbf{u} + \cos 2t\,\mathbf{w})\big]$$

Since A is real, real and imaginary parts are real solutions (Robinson Ch. 29). The real part is $\mathbf{x}_1$; the imaginary part is

$$\mathbf{x}_2(t) = e^{-t}(\sin 2t\,\mathbf{u} + \cos 2t\,\mathbf{w}) = e^{-t}\begin{pmatrix} \sin 2t + \cos 2t \\ \sin 2t + 2\cos 2t \end{pmatrix}$$

Step 2. Fundamental set

$\mathbf{x}_1(0) = \mathbf{u}, \mathbf{x}_2(0) = \mathbf{w}$, so the Wronskian at $t = 0$ is $W(0) = \det \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = 2 - 1 = 1 \neq 0$. $W(0) \neq 0$ is enough for independence (Notes, p. 9). Abel's formula for systems (**extension**) adds $W(t) = W(0)e^{(\operatorname{tr} A)t} = e^{-2t} \neq 0$ for all t. Hence the general real solution is

$$\mathbf{x}(t) = c_1 e^{-t}\begin{pmatrix} \cos 2t - \sin 2t \\ \cos 2t - 2\sin 2t \end{pmatrix} + c_2 e^{-t}\begin{pmatrix} \sin 2t + \cos 2t \\ \sin 2t + 2\cos 2t \end{pmatrix}, \qquad c_1, c_2 \in \mathbb{R}$$

Step 3. Initial condition

At $t=0$: $c_1\mathbf{u}+c_2\mathbf{w}=(1,0)^T$, i.e. $c_1+c_2=1$ and $c_1+2c_2=0$. Subtracting, $c_2=-1$, then $c_1=2$:

$$\mathbf{x}(t)=e^{-t}\begin{pmatrix}2(\cos 2t-\sin 2t)-(\sin 2t+\cos 2t)\\2(\cos 2t-2\sin 2t)-(\sin 2t+2\cos 2t)\end{pmatrix}=e^{-t}\begin{pmatrix}\cos 2t-3\sin 2t\\-5\sin 2t\end{pmatrix}$$

Step 4. Check by substitution

$\mathbf{x}(0)=(1,0)^T$. With $c=\cos 2t$, $s=\sin 2t$, the product rule gives $\mathbf{x}'=e^{-t}\big(-(c-3s)+(-2s-6c),\ 5s-10c\big)^T=e^{-t}(-7c+s,\ -10c+5s)^T$, and $A\mathbf{x}=e^{-t}\big(-7(c-3s)+4(-5s),\ -10(c-3s)+5(-5s)\big)^T=e^{-t}(-7c+s,\ -10c+5s)^T$. They agree.

KEY RESULTS

- $\mathbf{x}_2(t)=e^{-t}\begin{pmatrix}\sin 2t+\cos 2t\\ \sin 2t+2\cos 2t\end{pmatrix}$, $W(0)=1\neq 0$.
- $\mathbf{x}(t)=c_1\mathbf{x}_1(t)+c_2\mathbf{x}_2(t)$; IVP: $c_1=2$, $c_2=-1$.
- $\mathbf{x}(t)=e^{-t}\begin{pmatrix}\cos 2t-3\sin 2t\\ -5\sin 2t\end{pmatrix}$.

Part (c) · 1 point

Classification, spiralling and sense of rotation

Step 1. Type

The eigenvalues are complex with negative real part, so every solution is e^{-t} times a bounded π-periodic vector: it tends to $\mathbf{0}$ while winding around it. The origin is a stable focus (Robinson Ch. 31).

Step 2. Test vector

At $(1,0)$, on the positive x axis, $\mathbf{x}'=A(1,0)^T=(-7,-10)^T$: the y component is negative, so the solution crosses the positive x axis downwards. That is a **clockwise** rotation.

Step 3. Global confirmation

With θ the polar angle, $\theta'=\dfrac{xy'-yx'}{x^2+y^2}$. Here

$$xy'-yx'=x(-10x+5y)-y(-7x+4y)=-10x^2+12xy-4y^2=-2(5x^2-6xy+2y^2).$$

The quadratic form $5x^2-6xy+2y^2$ has matrix $\begin{pmatrix}5&-3\\-3&2\end{pmatrix}$ with $5>0$ and determinant $10-9=1>0$: it is positive definite. So $\theta'<0$ at every point except the origin: the rotation is clockwise everywhere. In the canonical plane J with $\omega=2>0$ rotates clockwise ($z_2'=-2z_1<0$ at $(1,0)$), and $\det P=1>0$

preserves orientation, which is consistent.

Step 4. Portrait in words

Trajectories are spirals that wind clockwise into the origin, one full turn every $T = 2\pi/\omega = \pi$ time units. They are not circles: they are images under P of logarithmic spirals of the $\mathbf{z}$ plane, so $\mathbf{u} = (1, 1)$ and $\mathbf{w} = (1, 2)$ are conjugate semi-diameters of the ellipses $V = \text{const}$ that the spirals cross.

KEY RESULTS

- $\rho = -1 < 0, \omega = 2 \neq 0$: **stable focus** (spiral sink); $\operatorname{tr} A = -2 < 0$, $\det A = 5 > 0$, $(\operatorname{tr} A)^2 - 4 \det A = -16 < 0$.
- $A(1, 0)^T = (-7, -10)^T$: **clockwise** rotation.
- $xy' - yx' = -2(5x^2 - 6xy + 2y^2) < 0$ off the origin.

Part (d) · 1 point

Decay of the canonical radius, return to the axis, Euclidean distance

Step 1. V is the canonical radius squared

$P^{-1}\mathbf{x} = \begin{pmatrix} 2x - y \\ -x + y \end{pmatrix}$, so $|P^{-1}\mathbf{x}|^2 = (4x^2 - 4xy + y^2) + (x^2 - 2xy + y^2) = 5x^2 - 6xy + 2y^2 = V$.

Step 2. Derivative along solutions

$V_x = 10x - 6y$, $V_y = -6x + 4y$, $x' = -7x + 4y$, $y' = -10x + 5y$. By the chain rule

$$\begin{aligned} \frac{dV}{dt} &= (10x - 6y)(-7x + 4y) + (-6x + 4y)(-10x + 5y) \\ &= (-70x^2 + 82xy - 24y^2) + (60x^2 - 70xy + 20y^2) = -10x^2 + 12xy - 4y^2 = -2V. \end{aligned}$$

(Equivalently $\frac{d}{dt}|\mathbf{z}|^2 = 2\mathbf{z}^T J \mathbf{z} = 2\rho|\mathbf{z}|^2$, because the skew part of J gives $\mathbf{z}^T \begin{pmatrix} 0 & \omega \\ -\omega & 0 \end{pmatrix} \mathbf{z} = 0$.)

Step 3. Integration

For $V \neq 0$ separate variables: $\displaystyle\int_{V(0)}^{V(t)} \frac{dV}{V} = \int_0^t (-2)\, d\tau$, so by Barrow's rule $[\ln V]_{V(0)}^{V(t)} = [-2\tau]_0^t$, i.e. $\ln V(t) - \ln V(0) = -2t$ and $V(t) = V(0)e^{-2t}$. For the IVP $V(0) = 5 \cdot 1^2 - 0 + 0 = 5$, so $V(t) = 5e^{-2t}$ and $\sqrt{V(t)} = \sqrt{5}\, e^{-t}$. Halving: $e^{-t_{1/2}} = \frac{1}{2}$, so $t_{1/2} = \ln 2 \approx 0.693$.

Step 4. Return to the positive x axis

$y(t) = -5e^{-t} \sin 2t = 0$ for $t = k\pi/2$. At $t = \pi/2$: $x = e^{-\pi/2} \cos \pi = -e^{-\pi/2} < 0$ (negative axis). At $t = \pi$: $x = e^{-\pi} \cos 2\pi = e^{-\pi} > 0$. So the first return is $t = \pi$, with $\mathbf{x}(\pi) = e^{-\pi}(1, 0)^T$ and ratio $|\mathbf{x}(\pi)|/|\mathbf{x}(0)| = e^{-\pi} \approx 0.0432$. This holds for every solution: the trigonometric factors have period $T = \pi$, so $\mathbf{x}(t + \pi) = e^{-\pi}\mathbf{x}(t)$.

Step 5. The Euclidean distance

$\frac{d}{dt}|\mathbf{x}|^2 = 2\mathbf{x}^T A\mathbf{x} = 2(-7x^2 + 4xy - 10xy + 5y^2) = 2(-7x^2 - 6xy + 5y^2)$. At $(0,1)$ it equals $10 > 0$: a solution through $(0,1)$ moves away from the origin at that instant. The Euclidean distance decays on average (by $e^{-\pi}$ per turn) but not monotonically; the quantity that decays monotonically, and exactly like e^{-t}, is $\sqrt{V} = |P^{-1}\mathbf{x}|$.

KEY RESULTS

- $V = |P^{-1}\mathbf{x}|^2 = 5x^2 - 6xy + 2y^2$, $\frac{dV}{dt} = -2V$, $V(t) = V(0)e^{-2t}$; for the IVP $V(t) = 5e^{-2t}$.
- $\sqrt{V}$ halves in $t_{1/2} = \ln 2 \approx 0.693$.
- Back on the positive x axis at $t = \pi$ (negative axis at $t = \pi/2$); $\mathbf{x}(\pi) = e^{-\pi}(1,0)^T$, ratio $e^{-\pi} \approx 0.0432$.
- No: $\frac{d}{dt}|\mathbf{x}|^2 = 2(-7x^2 - 6xy + 5y^2)$ is positive at $(0,1)$.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $t \to +\infty$: $\mathbf{x} \to \mathbf{0}$ like e^{-t}; $t \to -\infty$: the spirals grow without bound. If ρ were 0 the trajectories would be the closed ellipses $V = \text{const}$ (a centre); here they cut those ellipses inwards.
- **Checks:** $A\mathbf{x}_1 = \mathbf{x}_1'$ for all t; $\operatorname{tr} A = 2\rho$, $\det A = \rho^2 + \omega^2$; $P^{-1}AP = J$; the IVP solution satisfies the system and $\mathbf{x}(0) = (1,0)^T$; Abel: $W(t) = e^{-2t}$.
- **Common mistakes in the exam:** (1) reading $\mathbf{w}$ with the wrong sign: the course writes the real part as $\cos\omega t\,\mathbf{u} - \sin\omega t\,\mathbf{w}$, so $\mathbf{w} = +(1,2)$; (2) deciding the sense of rotation from $\omega > 0$ alone: it must be checked with a test vector (it is clockwise here, and it would flip if $\det P < 0$); (3) claiming that the Euclidean distance halves after $\ln 2$: only the canonical radius $|P^{-1}\mathbf{x}|$ does; (4) taking the return time to the same half-axis as π/ω instead of $2\pi/\omega$.

`1st Midterm · Unit 4 · Linear systems and phase portraits · 5 points · 30 min · Standard exam level`

P07 — Non-homogeneous planar system with a defective matrix: Jordan form, equilibrium shift, IVP and the improper node

Consider the system

$$x' = -x + y, \qquad y' = -x - 3y + 4.$$

(a) Write it as $\mathbf{x}' = A\mathbf{x} + \mathbf{f}$. Show that A has a repeated eigenvalue with only one independent eigenvector $\mathbf{v}$. Find a generalised eigenvector $\mathbf{w}$ with $(A - \lambda I)\mathbf{w} = \mathbf{v}$, build $P = [\mathbf{v} \mid \mathbf{w}]$, compute P^{-1} and check that $P^{-1}AP$ is the Jordan form.

(b) Find a particular solution by undetermined coefficients (a constant vector) and the general solution.

(c) Solve the initial value problem $x(0) = 3$, $y(0) = 0$ and check your answer.

(d) Classify the equilibrium and describe the phase portrait: invariant line, direction of tangency as $t \to \pm\infty$, sense of turning (test vector), nullclines, and the behaviour of the orbit of (c).

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknowns $x(t), y(t), t \in \mathbb{R}$. First-order, linear, constant coefficients, **non-homogeneous** with a constant forcing; solutions exist for all t and are unique given $\mathbf{x}(0)$.
- Structure of the general solution (superposition): $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$, with $\mathbf{x}_h$ the general solution of $\mathbf{x}' = A\mathbf{x}$.
- Since $\mathbf{f}$ is constant and 0 will not be an eigenvalue ($\det A \neq 0$), a constant $\mathbf{x}_p$ exists: it is the equilibrium, and $\mathbf{u} = \mathbf{x} - \mathbf{x}_p$ obeys the homogeneous system.

Phase 2. Canonical formulation

$$\mathbf{x}' = A\mathbf{x} + \mathbf{f}, \qquad A = \begin{pmatrix} -1 & 1 \\ -1 & -3 \end{pmatrix}, \qquad \mathbf{f} = \begin{pmatrix} 0 \\ 4 \end{pmatrix}.$$

Target frame: $\mathbf{u} = P\mathbf{z}$ with $P = [\mathbf{v} \mid \mathbf{w}]$, which turns $\mathbf{u}' = A\mathbf{u}$ into $\mathbf{z}' = J\mathbf{z}$, $J = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$.

Part (a) · 1.5 points

Repeated eigenvalue, generalised eigenvector and Jordan form

Step 1. Characteristic polynomial

$$\det(A-\lambda I) = (-1-\lambda)(-3-\lambda) - (1)(-1) = \lambda^2 + 4\lambda + 3 + 1 = \lambda^2 + 4\lambda + 4 = (\lambda+2)^2$$

so $\lambda = -2$ is a double root ($\operatorname{tr} A = -4 = 2\lambda$, $\det A = 4 = \lambda^2$).

Step 2. Eigenvectors

$N = A + 2I = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$. Row 1: $v_1 + v_2 = 0$; row 2: $-v_1 - v_2 = 0$, the same equation. The rank of N is 1, so the eigenspace is one-dimensional, spanned by $\mathbf{v} = (1,-1)^T$: A is not diagonalisable. Also $N^2 = \begin{pmatrix} 1-1 & 1-1 \\ -1+1 & -1+1 \end{pmatrix} = 0$, as the Cayley-Hamilton theorem predicts.

Step 3. Generalised eigenvector

$N\mathbf{w} = \mathbf{v}$: row 1 $w_1 + w_2 = 1$, row 2 $-w_1 - w_2 = -1$ (the same). Any solution works; take $w_2 = 0$, $w_1 = 1$: $\mathbf{w} = (1,0)^T$. It is not a multiple of $\mathbf{v}$, as required.

Step 4. Change of basis and Jordan form

$P = [\mathbf{v} \mid \mathbf{w}] = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}$, $\det P = 1\cdot 0 - 1\cdot(-1) = 1$, $P^{-1} = \frac{1}{1}\begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$.

$A\mathbf{v} = -2\mathbf{v} = (-2,2)^T$ and $A\mathbf{w} = (-1,-1)^T = \mathbf{v} - 2\mathbf{w}$, so $AP = \begin{pmatrix} -2 & -1 \\ 2 & -1 \end{pmatrix}$. Then

$$\begin{aligned} (P^{-1}AP)_{11} &= 0(-2) + (-1)(2) = -2, & (P^{-1}AP)_{12} &= 0(-1) + (-1)(-1) = 1, \\ (P^{-1}AP)_{21} &= 1(-2) + 1(2) = 0, & (P^{-1}AP)_{22} &= 1(-1) + 1(-1) = -2. \end{aligned}$$

$$P^{-1}AP = \begin{pmatrix} -2 & 1 \\ 0 & -2 \end{pmatrix} = J.$$

KEY RESULTS

- $p(\lambda) = \lambda^2 + 4\lambda + 4 = (\lambda+2)^2$, $\lambda = -2$ (double); $N = A + 2I = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$, rank 1, $N^2 = 0$.
- $\mathbf{v} = (1,-1)^T$, $\mathbf{w} = (1,0)^T$.
- $P = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}$, $\det P = 1$, $P^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$, $P^{-1}AP = \begin{pmatrix} -2 & 1 \\ 0 & -2 \end{pmatrix}$.

Part (b) · 1.5 points

Particular solution and general solution

Step 1. Undetermined constant vector

With $\mathbf{x}_p = (p,q)^T$ constant, $\mathbf{x}_p' = \mathbf{0}$ and the system gives $A\mathbf{x}_p = -\mathbf{f}$:

$$-p + q = 0, \qquad -p - 3q + 4 = 0.$$

The first gives $q = p$; substituting, $-4p + 4 = 0$, so $p = q = 1$. The solution is unique because $\det A = 4 \neq 0$. So $\mathbf{x}^* = (1,1)^T$ is the only equilibrium.

Step 2. Homogeneous part in the Jordan frame

Let $\mathbf{u} = \mathbf{x} - \mathbf{x}^*$; then $\mathbf{u}' = \mathbf{x}' = A\mathbf{x} + \mathbf{f} = A\mathbf{u} + (A\mathbf{x}^* + \mathbf{f}) = A\mathbf{u}$. With $\mathbf{u} = P\mathbf{z}$ (P constant): $\mathbf{z}' = P^{-1}AP\,\mathbf{z} = J\mathbf{z}$, i.e.

$$z_1' = -2z_1 + z_2, \qquad z_2' = -2z_2.$$

The second equation gives $z_2 = c_2 e^{-2t}$. The first becomes $z_1' + 2z_1 = c_2 e^{-2t}$; multiply by the integrating factor e^{2t}: $(e^{2t}z_1)' = e^{2t}z_1' + 2e^{2t}z_1 = c_2$ (product rule). Integrating, $e^{2t}z_1 = c_1 + c_2 t$, so $z_1 = (c_1 + c_2 t)e^{-2t}$.

Step 3. Back to the original frame

$\mathbf{u} = P\mathbf{z} = z_1\mathbf{v} + z_2\mathbf{w} = c_1 e^{-2t}\mathbf{v} + c_2 e^{-2t}(t\mathbf{v} + \mathbf{w})$, the formula of Robinson Ch. 30. Adding $\mathbf{x}^*$:

$$\mathbf{x}(t) = \begin{pmatrix} 1 + (c_1 + c_2 + c_2 t)e^{-2t} \\ 1 - (c_1 + c_2 t)e^{-2t} \end{pmatrix}.$$

The fundamental matrix $\Phi = \left[e^{-2t}\mathbf{v} \mid e^{-2t}(t\mathbf{v} + \mathbf{w})\right]$ has $\det \Phi = e^{-4t}\det[\mathbf{v} \mid t\mathbf{v} + \mathbf{w}] = e^{-4t}\det[\mathbf{v} \mid \mathbf{w}] = e^{-4t} \neq 0$, in agreement with Abel's formula for systems (**extension**) $W(t) = W(0)e^{(\operatorname{tr} A)t}$.

KEY RESULTS

- $\mathbf{x}_p = \mathbf{x}^* = (1,1)^T$.
- $\mathbf{x}(t) = \begin{pmatrix}1\\1\end{pmatrix} + c_1 e^{-2t}\begin{pmatrix}1\\-1\end{pmatrix} + c_2 e^{-2t}\left[t\begin{pmatrix}1\\-1\end{pmatrix} + \begin{pmatrix}1\\0\end{pmatrix}\right]$.
- In components: $x = 1 + (c_1 + c_2 + c_2 t)e^{-2t}$, $y = 1 - (c_1 + c_2 t)e^{-2t}$.

Part (c) · 1 point

Initial value problem x(0) = 3, y(0) = 0

Step 1. Constants

$x(0) = 1 + c_1 + c_2 = 3$ and $y(0) = 1 - c_1 = 0$. Hence $c_1 = 1$ and $c_2 = 2 - c_1 = 1$:

$$x(t) = 1 + (2+t)e^{-2t}, \qquad y(t) = 1 - (1+t)e^{-2t}.$$

Step 2. Check

By the product rule, $x' = e^{-2t} - 2(2+t)e^{-2t} = -(3+2t)e^{-2t}$ and $y' = -e^{-2t} + 2(1+t)e^{-2t} = (1+2t)e^{-2t}$. Right-hand sides:

$$-x + y = -1 - (2+t)e^{-2t} + 1 - (1+t)e^{-2t} = -(3+2t)e^{-2t},$$

$$-x - 3y + 4 = -1 - (2+t)e^{-2t} - 3 + 3(1+t)e^{-2t} + 4 = (1+2t)e^{-2t}.$$

Both equations hold, and $x(0) = 3, y(0) = 0$.

KEY RESULTS

- $c_1 = 1, c_2 = 1$.
- $x(t) = 1 + (2+t)e^{-2t}, y(t) = 1 - (1+t)e^{-2t}$.
- $x' = -(3+2t)e^{-2t}, y' = (1+2t)e^{-2t}$.

Part (d) · 1 point

Classification and phase portrait

Step 1. Type

A double negative eigenvalue with a one-dimensional eigenspace gives a stable improper node (Robinson Ch. 31): it lies on the parabola $(\mathrm{tr})^2 = 4\det$ of the trace-determinant plane, at the border between the nodes and the foci.

Step 2. Eigen-direction and tangency

The line $\mathbf{x}^* + s\mathbf{v}$, i.e. $x + y = 2$, is invariant ($c_2 = 0$: $\mathbf{u} = c_1 e^{-2t}\mathbf{v}$ moves straight to $(1,1)$). For $c_2 \neq 0$, $\mathbf{u} = e^{-2t}\big[(c_1 + c_2 t)\mathbf{v} + c_2\mathbf{w}\big]$ and the term $c_2 t\mathbf{v}$ dominates as $t \to \pm\infty$: the direction of $\mathbf{u}$ tends to $\pm\mathbf{v}$ at both ends. So every orbit enters $(1,1)$ tangent to the line $x + y = 2$, and far away (as $t \to -\infty$) it is again parallel to that line, on the other side: each orbit turns half a revolution.

Step 3. Sense of turning

With $\mathbf{u}' = A\mathbf{u}$: $u_1' = -u_1 + u_2$, $u_2' = -u_1 - 3u_2$, so

$$u_1 u_2' - u_2 u_1' = u_1(-u_1 - 3u_2) - u_2(-u_1 + u_2) = -u_1^2 - 2u_1u_2 - u_2^2 = -(u_1 + u_2)^2 \leq 0,$$

zero only on the invariant line. The polar angle around $(1,1)$ never increases: orbits turn **clockwise**. Test vector: at $(2,1)$, $\mathbf{u} = \mathbf{w} = (1,0)$, and $\mathbf{u}' = A\mathbf{w} = (-1,-1)^T$ points down, crossing the horizontal through $(1,1)$ downwards on its right: clockwise.

Step 4. Nullclines

$x' = 0$ on the line $y = x$ (vertical slopes there); $y' = 0$ on $x + 3y = 4$ (horizontal slopes). Both pass through $(1,1)$ and split the plane into four sectors where (x', y') keeps its signs; e.g. at $(3,0)$: $x' = -3 < 0$, $y' = 1 > 0$.

Step 5. The orbit of (c)

$u_1 + u_2 = (2 + t)e^{-2t} - (1 + t)e^{-2t} = e^{-2t} > 0$: the orbit never meets the invariant line and stays in the half-plane $x + y > 2$ (uniqueness forbids crossing an invariant line). For $t \geq 0$, $x' = -(3 + 2t)e^{-2t} < 0$ and $y' = (1 + 2t)e^{-2t} > 0$: it moves monotonically from $(3,0)$ to $(1,1)$, with $u_2/u_1 = -(1 + t)/(2 + t) \to -1$, i.e. arriving from the lower right tangent to $x + y = 2$. Backwards in time it crosses $y = 1$ at $t = -1$ and $x = 1$ at $t = -2$, and escapes to infinity parallel to $(-1,1)$.

KEY RESULTS

- $(1,1)$ is a **stable improper node**: $\operatorname{tr} A = -4 < 0$, $\det A = 4 > 0$, $(\operatorname{tr} A)^2 - 4\det A = 0$, one eigen-direction.
- Invariant line $x + y = 2$ (through $(1,1)$ along $\mathbf{v} = (1,-1)$); all other orbits are tangent to it at $(1,1)$ and parallel to it as $t \to -\infty$.
- $u_1 u_2' - u_2 u_1' = -(u_1+u_2)^2 \le 0$: orbits turn **clockwise**; test vector $A\mathbf{w} = (-1,-1)^T$ at $(2,1)$.
- Nullclines: $x' = 0$ on $y = x$; $y' = 0$ on $x + 3y = 4$. The orbit of (c) stays in $x + y > 2$; for $t \ge 0$ x decreases to 1 and y increases to 1.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $t \to +\infty$: $\mathbf{x} \to (1,1)$ like te^{-2t}, slower than a pure e^{-2t} because of the secular factor t. $t \to -\infty$: $|\mathbf{x}| \to \infty$. If $\mathbf{f} = \mathbf{0}$ the portrait is the same, centred at the origin.
- **Checks:** $P^{-1}AP = J$ with $\det P = 1$; $A\mathbf{x}^* + \mathbf{f} = \mathbf{0}$; the IVP solution satisfies both equations and the initial data; $\det \Phi = e^{-4t}$ matches Abel's formula for systems (extension).
- **Common mistakes in the exam:** (1) writing $\mathbf{x}_h = c_1 e^{-2t}\mathbf{v} + c_2 te^{-2t}\mathbf{v}$, which forgets $\mathbf{w}$ and is not a fundamental set; (2) solving $(A - \lambda I)\mathbf{w} = \mathbf{v}$ with the wrong sign of λ (using $A - 2I$); (3) trying a particular solution $\mathbf{c}\,t$ or $\mathbf{c}\,e^{-2t}$ for a constant forcing: a constant vector suffices since $\det A \neq 0$; (4) drawing a star node (all straight lines), or claiming orbits cross the eigenline.